

A
Reserch Project
On

“An Analytics Study on Customer Personalization & Add-On Buying Behaviour in Food Delivery Apps.”

SUBMITTED TO



SAVITRIBAI PHULE PUNE UNIVERSITY

In the partial Fulfilment of the requirement for the award of the degree of Master of Business Administration (MBA)

Submitted By

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DECLARATION

I hereby declare that the Field Project entitled **“An Analytics Study on Customer Personalization & Add-On Buying Behaviour in Food Delivery Apps.”**

Written and submitted by me to the University of Pune, in partial fulfillment of the requirements for the award of degree of Master of Business Administration under the guidance of Prof. Miss. P. S. Salbande , this my original work and the conclusion drawn there in are based on the material Collected by myself .

PLACE : Sangamner

(Mr. Kale Vaibhav Dnyandev)

DATE:

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I wish to express a special thanks to all teaching and non-teaching staff members of Amrutvahini Institute of Management and Business Administration, Sangamner for their continuous support. I would like to acknowledge all my family members, relatives and friends for their help and encouragement.

Place: Sangamner

Date:

Signature:

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INDEX

Sr. No.	Particulars	Page No.
1	Executive Summary	1 - 2
2	Introduction	3 - 4
3	Industry Profile	5 - 6
4	Company Profile	7 - 10
5	Product Profile	11 -13
6	Theoretical Background	14 - 15
7	Literature Review	16 - 18
8	Problem Statement	19
9	Research Methodology a. Statement Of Research Proposal b. Rationale of the study c. Objectives of the Study d. Hypotheses of the Study e. Types of Research Design f. Sources of Data Collection g. Sample Design h. Need and Significance of the Study i. Statistical Tool j. Limitations of the Study	20 - 23
10	Data Analysis and Interpretation	24 - 50
11	Results	51
12	Findings	52 - 55
13	Conclusion	56
14	Suggestions	57 - 58
15	Discussion	59 - 60
16	Annexure	61 – 69

SUMMARY

This study examines the role of personalization in influencing customer behavior, with a specific focus on add-on buying behaviour in food delivery applications such as Zomato and Swiggy. The rapid growth of digital technologies and the increasing adoption of smartphones have significantly transformed the food delivery industry in India. Food delivery platforms are no longer limited to facilitating orders; instead, they operate as data-driven systems that analyze user behaviour, predict preferences, and deliver personalized experiences. In this context, personalization has emerged as a key strategy for enhancing customer engagement and increasing revenue through additional purchases.

The primary objective of the study is to analyze how personalization strategies affect add-on buying behaviour and to identify the key factors that influence customer purchasing decisions. Add-on items such as beverages, desserts, and side dishes contribute significantly to increasing the Average Order Value (AOV), making them an important focus area for food delivery platforms. Despite the widespread use of personalization techniques, it remains unclear whether these strategies effectively influence customer behaviour, particularly in the Indian market where price sensitivity is high and competition is intense.

The study is based on both primary and secondary data. Primary data was collected through a structured questionnaire administered to 118 respondents who actively use food delivery applications. The survey captured detailed information related to demographic characteristics, ordering frequency, spending patterns, platform preferences, and responses to various personalization features such as recommendations, discounts, push notifications, and combo offers. Secondary data was collected from research journals, company reports, and published studies to provide theoretical support and industry insights. The research adopts a quantitative approach using descriptive and causal research designs, supported by statistical tools such as percentage analysis, mean analysis, and chi-square tests.

The findings of the study reveal that personalization has a significant but supportive role in influencing customer behaviour. A majority of respondents indicate that personalized recommendations, notifications, and suggestions make the ordering process easier and influence their choices. However, this positive perception does not always translate into actual add-on purchases. A considerable proportion of users report that they rarely purchase add-on items, indicating that personalization alone is not sufficient to drive conversion.

One of the most important findings of the study is that price sensitivity plays a dominant role in shaping customer decisions. Discounts and promotional offers emerge as the primary factors influencing food ordering behaviour, with a large percentage of respondents identifying them as the main reason for using food delivery applications. Personalized discounts further increase the likelihood of purchase, highlighting the importance of combining personalization with pricing strategies.

The study also identifies combo deals and bundled offers as the most effective drivers of add-on purchases. Customers perceive combo offers as providing better value and convenience, which reduces decision-making effort and encourages impulse buying. Features such as “frequently bought together” and checkout-stage recommendations are particularly effective, as they influence customer decisions at the final stage of the purchasing process.

Another key finding is that add-on purchases are not habitual and require active triggers. A significant number of users purchase add-ons only occasionally or rarely, indicating that platforms must actively encourage such behaviour through targeted strategies. The preference for add-on items is also uneven, with beverages and side dishes being more popular than desserts, suggesting that customers prefer practical and complementary items rather than indulgent ones.

The analysis further highlights the role of demographic and economic factors in influencing customer behaviour. Income level has a significant impact on add-on purchasing decisions, with higher-income users showing greater willingness to spend on additional items. Ordering frequency also plays an important role, as users who order more frequently tend to generate higher average order values. These findings emphasize the importance of customer segmentation in designing effective personalization strategies.

In terms of platform comparison, the study finds that Zomato has a slight edge in user preference, but a large proportion of users actively switch between Zomato and Swiggy. This indicates low platform loyalty and suggests that customers are highly sensitive to offers, pricing, and user experience. Additionally, users perceive both platforms as offering similar levels of personalization, indicating limited differentiation in recommendation strategies.

The findings from secondary data analysis support the results obtained from primary data. While personalization improves customer engagement and purchase intention, its impact is significantly enhanced when combined with pricing strategies such as discounts and combo offers. The data also reveals that higher order frequency and income levels contribute to increased spending and add-on purchases, reinforcing the importance of targeting specific customer segments.

Overall, the study concludes that personalization plays an important but secondary role in influencing add-on buying behaviour. While it enhances user experience and supports decision-making, actual purchase behaviour is primarily driven by perceived value, pricing incentives, and convenience. Therefore, food delivery platforms should focus on integrating personalization with effective pricing strategies, such as targeted discounts and bundled offers, to maximize add-on purchases and revenue.

The study provides valuable insights for both academic and practical purposes. From a business perspective, it highlights the need for data-driven strategies that combine personalization with pricing and behavioral triggers. From an academic perspective, it contributes to the limited research available on add-on buying behaviour in the Indian context and provides a foundation for further studies in this area.

Introduction

The rapid advancement of digital technologies and the widespread adoption of smartphones have significantly transformed consumer behavior across various industries, particularly in the food delivery sector. In India, the growth of online food delivery platforms such as Zomato and Swiggy has reshaped how consumers access food services, making ordering more convenient, faster, and highly personalized. These platforms have evolved from simple ordering interfaces into sophisticated, data-driven ecosystems that utilize business analytics and artificial intelligence to understand and influence customer behavior.

The increasing pace of urbanization, changing lifestyles, and rising work commitments have contributed to a growing demand for convenient food solutions. Consumers, especially in urban and semi-urban areas, prefer food delivery services due to time constraints, variety, and ease of access. As a result, food delivery applications have become an integral part of daily life, particularly among younger and working populations. This shift in consumption patterns has encouraged companies to adopt advanced technological solutions to remain competitive and meet evolving customer expectations.

One of the most critical developments in this domain is the use of personalization strategies. Food delivery platforms collect and analyze large volumes of user data, including browsing history, order patterns, preferences, and behavioral interactions. This data is used to provide personalized recommendations, targeted promotions, and push notifications aimed at enhancing user experience and increasing customer engagement. Personalization not only simplifies decision-making for users but also creates opportunities for platforms to influence purchasing behavior through data-driven insights.

A key outcome of personalization is its impact on add-on buying behaviour. Add-on purchases refer to additional items such as beverages, desserts, or side dishes that customers include along with their primary order. These purchases play a significant role in increasing the Average Order Value (AOV) and overall profitability of food delivery platforms. From a business perspective, encouraging add-on purchases is more cost-effective than acquiring new customers, making it an important focus area for revenue optimization strategies.

However, despite the widespread implementation of personalization techniques, it remains unclear whether these strategies effectively drive add-on purchases. Customers often exhibit price-sensitive behavior and may prioritize discounts and offers over personalized recommendations. Additionally, user preferences vary significantly based on factors such as income, ordering frequency, and demographic characteristics. This creates challenges for platforms in designing effective personalization strategies that can consistently influence customer decisions.

Existing literature suggests that personalization improves customer engagement and purchase intention by delivering relevant and timely recommendations. Studies also indicate that behavioral factors such as perceived value, convenience, and ease of use play an important role in shaping customer decisions. However, there is limited empirical research in the Indian context that specifically examines the relationship between personalization and add-on buying behaviour. Moreover, the relative effectiveness of different factors such as discounts, combo offers, and recommendation systems in influencing add-on purchases remains underexplored. This study addresses these gaps by analyzing how personalization strategies influence add-on buying behaviour in food delivery applications. It focuses on identifying key determinants such as pricing strategies, recommendation systems, customer preferences, and behavioral triggers that impact purchasing decisions. The study also compares customer behavior across major platforms like Zomato and Swiggy to understand differences in user perception and platform effectiveness.

The research adopts a data-driven approach by combining primary data collected through a structured questionnaire with secondary data from academic and industry sources. This approach allows for a comprehensive understanding of both theoretical concepts and real-world customer behavior. By applying statistical analysis and examining behavioral patterns, the study aims to provide meaningful insights into the effectiveness of personalization strategies. The rationale behind this study lies in the growing importance of personalization as a competitive strategy in the digital economy. As the food delivery market becomes increasingly saturated, platforms must move beyond generic marketing approaches and adopt targeted, data-driven strategies to retain customers and increase revenue. Understanding the factors that influence add-on purchases can help businesses design more effective recommendation systems, optimize pricing strategies, and improve overall customer experience.

Furthermore, this study contributes to the field of business analytics by demonstrating how data-driven insights can be applied to understand and influence consumer behavior. It highlights the intersection of analytics, marketing, and consumer psychology in shaping digital purchasing decisions. The findings of this research are expected to provide practical implications for food delivery platforms, marketers, and business analysts, while also contributing to academic literature by addressing gaps in existing research.

Industry Profile

1. Overview

The food delivery industry in India has emerged as one of the fastest-growing sectors, driven by digitalization, urbanization, and changing consumer lifestyles. App-based platforms have transformed food ordering into a convenient, real-time service supported by data analytics and technology. These platforms not only facilitate ordering but also influence customer behavior through personalized recommendations and offers.

- Key Platforms: Zomato, Swiggy
- Industry Type: Online Food Delivery & Aggregator Services
- Growth Period: Major expansion post-2015 with rapid digital adoption
- Coverage: Pan-India (Urban + Tier 2 & 3 cities)
- Core Services: Food ordering, delivery logistics, restaurant aggregation

2. Business Model & Revenue

- **Aggregator Model:** Connects customers, restaurants, and delivery partners through a digital platform.
- **Revenue Streams:**
 - i. Commission from restaurants (per order)
 - ii. Delivery charges from customers
 - iii. Advertising & promotional listings
 - iv. Subscription services (Zomato Gold, Swiggy One)
- **Data-Driven Model:** Uses analytics for personalization, pricing, and recommendations.

3. Market Position & Competitors

- Market Leaders: Zomato and Swiggy dominate the Indian market.
- Competitors: Amazon Food (limited), ONDC platforms, local delivery services
- Key Strengths:
 - i. Strong logistics and delivery network
 - ii. Advanced personalization using AI & analytics
 - iii. Wide restaurant partnerships and variety
 - iv. High customer engagement through offers & discounts

4. Growth Strategies

- Expansion: Increasing penetration in Tier 2 & Tier 3 cities
- Personalization: AI-based recommendations and targeted offers
- Subscription Models: Loyalty programs for customer retention
- Cloud Kitchens: Expansion of delivery-only restaurant models
- Technology Integration: Use of machine learning, real-time tracking, and smart suggestions

5. Challenges

- High Competition: Intense rivalry between platforms and price wars
- Profitability Issues: High delivery and operational costs
- Discount Dependency: Heavy reliance on offers to attract customers
- Logistics Management: Timely delivery and workforce management

6. Future Outlook

- AI & Personalization: More advanced recommendation systems
- Growth in Add-On Sales: Focus on increasing average order value
- Rural Expansion: Entry into smaller cities and towns
- Sustainability: Eco-friendly packaging and efficient delivery models
- Integration with ONDC: Increasing competition and open-network growth

Company Profile

1. Overview



Zomato

Founded:	2008
Headquarters:	Gurugram, Haryana, India
Founder:	Deepinder Goyal
Industry:	Food Delivery & Restaurant Aggregator
Key Services:	Food delivery, dining, reviews, grocery (Blinkit)



Swiggy

Founded	2014
Headquarters	Bengaluru, Karnataka, India
Founder	Sriharsha Majety, Nandan Reddy
Industry	Food Delivery & Logistics Platform
Key Services	Food delivery, Instamart (grocery), Genie (pickup/drop)

2. Business Model & Revenue

Zomato

- Commission from restaurants
- Delivery charges from customers
- Advertising & promoted listings
- Subscription: Zomato Gold
- Revenue diversification via Blinkit (quick commerce)

Swiggy

- Commission-based revenue
- Delivery & surge charges
- Subscription: Swiggy One
- Revenue from Instamart & logistics services
- Strong logistics-focused model

3. Market Position & Strengths

Zomato

- Strong brand recognition
- Leader in restaurant discovery & reviews
- Integrated ecosystem (delivery + dining + grocery)
- Data-driven personalization

Swiggy

- Strong delivery network & faster logistics
- High customer satisfaction in delivery experience
- Better operational efficiency
- Focus on convenience & speed

4. Personalization & Technology

Zomato

- AI-based recommendations
- Personalized restaurant listings
- Dynamic pricing & offers
- “Frequently ordered” suggestions

Swiggy

- Smart recommendations based on behavior
- Real-time tracking & predictive delivery time
- Personalized push notifications
- “Frequently bought together” features

5. Add-On Purchase Strategy

Zomato

- Combo deals & discounts
- Add-on suggestions during checkout
- Cross-selling via “recommended with this item”

Swiggy

- Strong UI for add-ons in cart
- Bundled meal offers
- Behavioral nudges like “complete your meal”
- Higher focus on impulse buying triggers

6. Challenges

Zomato

- Profitability pressure
- High competition
- Dependency on discounts
- Managing multiple business verticals

Swiggy

- High operational costs
- Logistics complexity
- Price competition with rivals
- Maintaining delivery efficiency

7. Future Outlook

Zomato

- Expansion in quick commerce (Blinkit)
- Advanced AI personalization
- Growth in premium dining services

Swiggy

- Expansion of Instamart
- Hyperlocal delivery ecosystem
- Increased focus on speed & convenience

Product Profile



Paneer Butter Masala with Naan



Vegetable Biryani



Masala Dosa with Sambar & Chutney



Pizza (Cheese / Margherita Style)



Veg Burger



Pav Bhaji



Butter Chicken



Chicken Wrap / Shawarma Roll



Cheese Pizza



Chicken Burger (Crispy Burger)



Seekh Kebab



Chicken Tikka



**Chocolate Milkshake / Oreo Shake
(Desserts)**



Soft Drinks (Beverages)



Chocolate Brownie Sundae (Desserts)



French Fries Ketchup (Sides)



Fried Samosa (Veg Starter)



Mango Milkshake (Beverages)

Theoretical Background

An Analytics Study on Customer Personalization & Add-On Buying Behaviour in Food Delivery Apps

Category	Concepts / Theories	Application (Food Delivery Apps)
Data Analytics Framework	Descriptive, Diagnostic, Predictive, Prescriptive Analytics	Tracking past orders; identifying reasons for add-ons; predicting customer preferences; recommending combos and upsell items
Personalization Systems	Content-Based Filtering, Collaborative Filtering, Hybrid Recommendation Systems	Suggesting relevant add-ons like drinks, desserts, or sides based on user history and similar users
Consumer Behavior	Theory of Planned Behavior, Impulse Buying, Behavioral Economics (Anchoring, Decoy Pricing, Loss Aversion)	Influencing customers to add items through offers like “Add Coke @ ₹20” or combo deals
Market Segmentation & Insights	Demographic, Behavioral, Psychographic Segmentation; Cohort Analysis; Customer Data Platforms (CDPs)	Grouping users (students, families, frequent buyers) and offering personalized deals and recommendations
Predictive Modeling	Regression, Classification, Clustering (K-means), Recommendation Algorithms	Forecasting demand; identifying users likely to accept add-ons; optimizing targeting strategies
Marketing & Strategy Models	4Ps (Product, Price, Place, Promotion), SWOT Analysis, Porter’s Five Forces	Designing combo meals, pricing add-ons, promoting offers via app notifications, and analyzing competition
UI/UX & Choice Architecture	Nudge Theory, Default Options, Choice Architecture	“Frequently Bought Together”, pre-selected add-ons, and visually highlighted offers to increase conversions

Category	Concepts / Theories	Application (Food Delivery Apps)
Visualization & BI	Dashboards, Heatmaps, A/B Testing	Tracking user behavior, analyzing click patterns, and optimizing add-on placement in the app
Modern Analytics Trends	Big Data Analytics, Real-Time Analytics, AI & Machine Learning	Handling large-scale user data and providing real-time personalized recommendations
Business Impact	Customer Lifetime Value (CLV), Average Order Value (AOV), Retention Models	Increasing revenue through add-ons, improving customer retention, and enhancing user experience

Literature Review

Prasetyo, H., Nugroho, A., & Putri, S. (2026)

They examined consumer purchasing behavior on food delivery platforms such as GoFood, GrabFood, and ShopeeFood. Using descriptive statistics and ANOVA on survey data from 400 users, they found that ease of use significantly influences consumer buying behaviour, while promotional activities and food variety create differences in purchasing patterns. They also identified that social influence and perceived price value play important roles in shaping online food ordering decisions. This study supports the idea that personalized promotions and tailored recommendations can influence add-on purchases and consumer engagement in food delivery apps. The findings also suggest that user-friendly interfaces and targeted marketing strategies can strengthen customer loyalty and encourage repeated purchasing behaviour over time.

Shorbaji, M. F. (2025)

He conducted a systematic review of more than 50 studies on AI-enabled mobile food-ordering applications. The research found that AI technologies such as personalized recommendations, chatbots, and context-aware suggestions improve customer experience and application engagement. However, the study also noted that excessive personalization, repetitive suggestions, and lack of pricing transparency may reduce consumer trust. This research highlights how personalization strategies can increase basket size and encourage add-on purchases by improving recommendation relevance. Additionally, the study emphasizes the importance of balancing personalization with transparency to maintain long-term user satisfaction and platform credibility.

Zhao, L. (2025)

He studied how personalized recommendations influence consumer purchase intentions using Self-Determination Theory. Through experimental research, the study found that personalized suggestions satisfy psychological needs such as competence and relatedness, which increases engagement and purchase intention. The results showed that consumers are more likely to respond to personalized offers and upselling prompts when recommendations align with their preferences. This study provides theoretical support for understanding how personalization affects consumer behaviour in food delivery platforms. It also highlights that motivational factors play an important role in shaping consumer responses to recommendation systems.

Ahn, J., & Kwon, H. (2025)

They investigated the impact of AI-driven personalized recommendations on consumer purchasing behaviour using the Stimulus-Organism-Response (S-O-R) framework. Using structural equation modeling on survey data, they found that personalization positively affects purchase intention both directly and through perceived value. They also discovered that transparency in recommendation systems increases consumer trust. This research suggests that personalized recommendation systems can encourage add-on purchases when consumers perceive higher value and trust in the platform. Furthermore, the study emphasizes the role of transparency in strengthening customer confidence toward AI-driven decision support systems.

Consumer Buying Behavior Towards Online Food Delivery Apps (2025)

This study examined the key factors influencing consumer behaviour in online food delivery applications, including website design, security, trust, and customer reviews. Using a mixed-method approach involving surveys and case studies, the research found that both technological features and psychological trust significantly influence purchase intentions and repeat usage. The study also emphasized that customized user interfaces and personalized content reduce decision-making effort and improve user engagement. These findings indicate that personalization can increase consumer confidence and encourage additional purchases. The research further highlights the importance of user-friendly platforms in enhancing the overall digital food ordering experience.

Impact of Personalized Advertisements on Food Delivery Decisions (2024)

This study investigated how personalized advertisements influence consumer decision-making in food delivery apps in the Delhi-NCR region. Using surveys and focus group discussions, the research found that targeted advertisements based on past ordering behavior significantly increase click-through rates and order values. Consumers reported that relevant and timely promotions strongly influence their purchasing decisions. This research demonstrates that data-driven advertising strategies can effectively stimulate add-on purchases and enhance customer engagement. It also indicates that personalized promotional content helps companies better understand consumer preferences and improve marketing effectiveness.

Li, X., Wang, Y., & Zhang, H. (2024)

They analyzed the impact of machine learning-based personalization algorithms on consumer decision-making in mobile commerce platforms. Using transaction data and A/B testing experiments, the study found that personalized recommendation ranking, bundle suggestions, and “frequently bought together” prompts significantly increase basket size and cross-selling opportunities. The research also noted that excessive personalization may reduce consumer autonomy and trust. This study provides an analytical framework for understanding how predictive analytics can increase add-on purchases in digital food ordering platforms. It also highlights the importance of balancing algorithmic recommendations with user control.

Verhoef, P. C., Kannan, P. K., & Inman, J. J. (2023)

They explored how advanced data analytics supports omnichannel personalization strategies in digital commerce. The study found that integrating customer transaction data, behavioral patterns, and contextual signals allows platforms to create highly relevant product recommendations. These personalized experiences reduce consumer search effort and increase engagement, which leads to higher purchase likelihood and cross-selling opportunities. This research highlights the importance of data-driven personalization in increasing customer lifetime value and purchase behaviour. The authors also emphasize the role of integrated digital platforms in delivering consistent personalized experiences.

Sánchez, C. N., et al. (2023)

They developed a recommendation system for food delivery applications based on the number of past orders. Using near-neighbor algorithms and real transaction data, the study demonstrated that order history alone can effectively identify consumer preferences and generate relevant recommendations. The results showed that personalized recommendations reduce choice overload and improve user satisfaction. This approach highlights how analytics based on transaction history can encourage add-on purchases by presenting relevant food suggestions. The research further suggests that simple recommendation models can still produce meaningful personalization results.

Hallikainen, H., et al. (2022)

They examined the consequences of personalized product recommendations in digital commerce. Their research found that algorithm-driven recommendations influence both product selection and purchase depth by reducing search costs and increasing perceived relevance. The study distinguishes between transaction personalization and decision personalization, showing how both strategies guide consumer behaviour. These insights suggest that personalization algorithms can increase attach rates and add-on purchases in digital platforms such as food delivery apps. The study also highlights the importance of ethical data usage and consumer trust in recommendation systems.

Tandon, A., et al. (2021)

They studied why consumers purchase from food delivery apps using the Theory of Consumption Values. Surveying 355 users in the United States, the study found that functional values such as convenience and ease of use significantly influence consumer attitudes and purchase intentions. The results indicate that positive attitudes toward food delivery platforms mediate the relationship between perceived value and actual purchase behaviour. This research suggests that personalized services enhancing convenience can increase order frequency and additional purchases. The study also highlights the importance of service efficiency in maintaining customer satisfaction.

A Study of Consumer Behaviour Towards Online Food Delivery (2019)

This study investigated factors influencing consumer preferences for online food delivery services through questionnaire-based research. It identified promotions, application usability, brand reputation, and service reliability as key determinants influencing consumer choice of delivery platforms. The study highlights that promotional strategies and user experience significantly affect consumer ordering behaviour. These early findings provide foundational insights into consumer behaviour before the widespread adoption of advanced personalization technologies. The research also demonstrates how marketing incentives influence consumer platform preference and ordering frequency.

Problem Statement

Category	Problem / Task	Description
Problem Identification	Low Add-On Conversion	Customers on platforms like Zomato and Swiggy often ignore add-on suggestions such as drinks, desserts, and sides
Key Issues	Ineffective Personalization	Recommendations are sometimes irrelevant, poorly timed, or not attractive enough to influence purchase decisions
Customer Behavior Gap	Variation in Preferences	Differences in customer needs, price sensitivity, and ordering habits make it difficult to design universal personalization strategies
Research Focus	Personalization Impact	Study how personalized recommendations influence add-on buying behavior
Data Collection Task	Primary Data Collection	Collect data through structured questionnaires (Google Forms) from users of food delivery apps
Customer Analysis	Segmentation	Classify customers based on demographics, behavior, and preferences
Analytical Approach	Data Analysis Techniques	Use descriptive statistics, charts, and comparative analysis to identify patterns
Behavioral Factors Study	Purchase Drivers	Analyze impact of discounts, combo offers, pricing, and visual cues on add-on purchases
System Evaluation	Recommendation Effectiveness	Evaluate how well current personalization and recommendation systems perform
Improvement Task	Strategy Development	Suggest improvements using data analytics and predictive modeling techniques
Expected Outcome	Business Insights	Increase average order value (AOV), improve customer satisfaction, and optimize personalization strategies

Research Methodology

a. Statement Of Research Proposal

The present study titled “An Analytics Study on Customer Personalization & Add-On Buying Behaviour in Food Delivery Apps” seeks to examine the influence of personalization strategies on customer add-on purchase behaviour in digital food delivery platforms. Food delivery applications such as Zomato and Swiggy extensively utilize data-driven personalization techniques including customized recommendations, targeted offers, and push notifications to enhance user engagement and increase average order value.

Add-on products such as beverages, desserts, and side dishes contribute significantly to incremental revenue. However, limited empirical research has specifically analyzed the impact of personalization on add-on buying behaviour, particularly in the Indian context.

This study aims to analyze the effect of personalization on add-on purchases, compare customer preferences between major platforms, and assess determinants influencing add-on spending patterns. The research adopts a quantitative and analytical approach using primary data collected through a structured questionnaire and applies statistical tools to test the stated hypotheses.

b. Rationale of the study

The increasing competition in the food delivery industry has compelled platforms to focus on revenue optimization through personalization strategies. While customer acquisition remains important, enhancing average order value through add-on purchases has become a key profitability driver.

Personalization features are designed to influence purchasing decisions by offering relevant recommendations and targeted promotions. However, there is limited localized research examining whether these personalization mechanisms significantly affect add-on buying behaviour.

The present study is necessary to provide empirical evidence on the relationship between personalization and incremental purchasing behaviour. It will assist food delivery platforms in refining recommendation systems and promotional strategies while contributing to academic literature by addressing the gap in platform-level comparative analysis of add-on buying behaviour.

c. Objectives of the Study

- 1 : To analyze personalization effects on add-on purchases.
- 2 : To compare customer add-on preferences between Zomato and Swiggy.
- 3 : To evaluate personalization features of food delivery apps.
- 4 : To assess determinants of customers’ add-on spending patterns.

5 : To examine add-on buying behavior across customer segments.

d. Hypotheses of the Study

1 : To compare customer add-on preferences between Zomato and Swiggy.

- **H₀:** There is no significant difference in add-on preferences between Zomato and Swiggy.
- **H₁:** There is a significant difference in add-on preferences between Zomato and Swiggy.

2 : To evaluate personalization features of food delivery apps.

- **H₀:** Personalization features do not significantly influence add-on buying decisions.
- **H₁:** Personalization features significantly influence add-on buying decisions.

3: To assess determinants of customers' add-on spending patterns.

- **H₀:** Determinants of add-on spending have no significant effect.
- **H₁:** Determinants of add-on spending have a significant effect.

e. Types of Research Design

Descriptive Research Design :

Descriptive research is used to analyze customer preferences, perceptions, and patterns of add-on purchases in food delivery applications. It helps describe behavioral trends using primary data collected from users of Zomato and Swiggy.

Causal (Explanatory) Research Design:

Causal research is applied to examine the cause-and-effect relationship between personalization strategies and add-on buying behaviour. It is used to test the hypotheses and determine whether personalization significantly influences add-on purchases.

f. Sources of Data Collection

The study is based on both **primary and secondary data**:

Primary Data :

Primary data was collected through a structured questionnaire administered to users of food delivery apps. It captured information on personalization features, add-on buying behaviour, and spending patterns. This provided direct insights into customer perceptions and decision-making.

Secondary Data :

Secondary data was gathered from research journals, company reports, websites, and published studies related to food delivery platforms and consumer behaviour. It helped build the theoretical foundation and understand industry trends. This supported the analysis and interpretation of primary data findings.

g. Sample Design

The sample design is structured to obtain relevant data aligned with the objectives and hypotheses of the study.

- a) **Sample Population:** Customers using food delivery applications.
- b) **Sampling Frame:** Active users of food delivery apps during the study period.
- c) **Sample Area:** Sangamner city.
- d) **Sample Size:** 100–120 respondents.
- e) **Sample Technique:** Convenience sampling method.
- f) **Sample Instrument:** Structured questionnaire consisting of close-ended and Likert-scale questions related to personalization and add-on buying behaviour.

h. Need and Significance of the Study

The study is significant as it helps food delivery platforms understand customer preferences for add-on products and the role of personalization in influencing purchase decisions. It provides insights useful for marketers, app developers, and business analysts to design better recommendation strategies and improve customer satisfaction and revenue.

i. Statistical Tool

The following statistical tools are used for data analysis:

- Percentage analysis
- Mean and ranking analysis
- Chi-square test
- Graphs and charts
- Data analysis is performed using **MS Excel**

j. Limitations of the Study

The present study is subject to certain limitations that should be acknowledged while interpreting the findings. First, the research is based on a relatively small sample size, which may limit the generalizability of the results to the broader population of food delivery app users. A larger sample could provide more comprehensive insights and improve statistical reliability. Second, the data collected relies primarily on customer perceptions and self-reported responses. Such responses may be influenced by personal bias, memory errors, or social desirability tendencies, which could affect the accuracy of the findings.

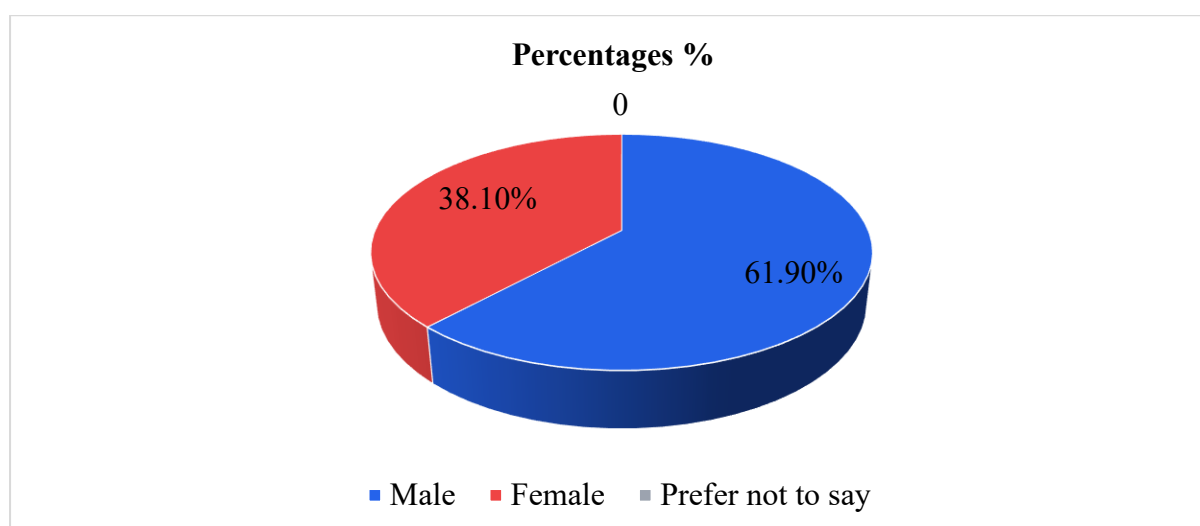
Additionally, time constraints posed challenges in conducting an extensive and in-depth analysis, thereby limiting the scope of data collection and advanced statistical exploration. Finally, the study focuses only on selected food delivery platforms, which restricts the applicability of the conclusions to other emerging or regional platforms. Despite these limitations, the study provides valuable insights into personalization and add-on buying behaviour within the specified research context.

DATA ANALYSIS AND INTERPRETATION

i). Gender

Data Table 1 : 118 Responses

Sr. No	Parameters	Responses	Percentages %
1	Male	73	61.90%
2	Female	45	38.10%
3	Prefer not to say	0	0
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The survey data indicates that 61.90% of respondents are male, while 38.10% are female. No respondents selected the “Prefer not to say” option. This shows that male respondents form the majority, although female participation is also significant.

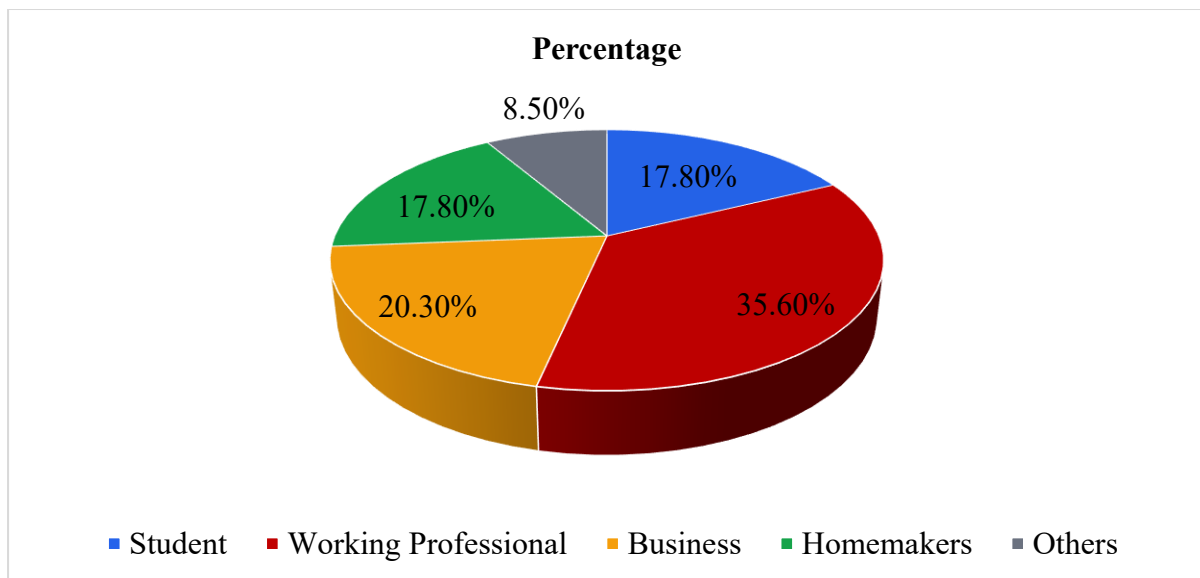
Data Interpretation :

The findings suggest that the respondent base is male-dominated but relatively balanced, with a considerable proportion of female participants. This indicates that food delivery applications are used by both genders, though slightly more by males. The absence of responses under “Prefer not to say” reflects that participants were comfortable disclosing their gender.

ii). Occupation.

Data Table 2 : 118 Responses

Sr. No	Parameters	Responses	Percentages %
1	Student	44	17.80%
2	Working Professional	30	35.60%
3	Business	18	20.30%
4	Homemakers	18	17.80%
5	Others	8	8.50%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The occupational distribution shows that working professionals constitute the largest group (35.6%), followed by students (17.8%) and homemakers (17.8%). Business respondents account for 20.3%, while others represent 8.5%, indicating a diverse but employment-skewed sample.

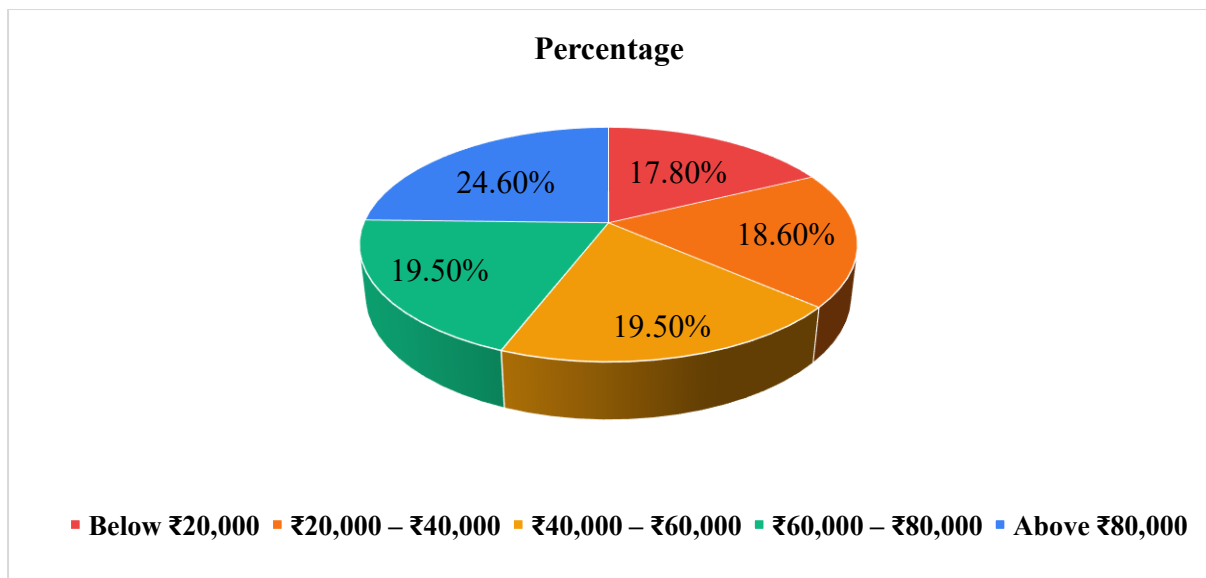
Data Interpretation :

The dominance of working professionals suggests higher reliance on convenience-driven services like food delivery. Students and homemakers contribute moderately, reflecting varied usage patterns, while business users indicate stable purchasing power, highlighting broad adoption across different occupational segments.

iii). Incomes.

Data Table 3 : 118 Responses

Sr. No	Income Range	Respondents	Percentage
1	Below ₹20,000	21	17.8%
2	₹20,000 – ₹40,000	22	18.6%
3	₹40,000 – ₹60,000	23	19.5%
4	₹60,000 – ₹80,000	23	19.5%
5	Above ₹80,000	29	24.6%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The income distribution indicates that respondents earning above ₹80,000 form the largest group (24.6%). Middle-income groups between ₹40,000–₹80,000 are evenly represented (19.5% each), while lower-income categories contribute comparatively smaller but consistent proportions.

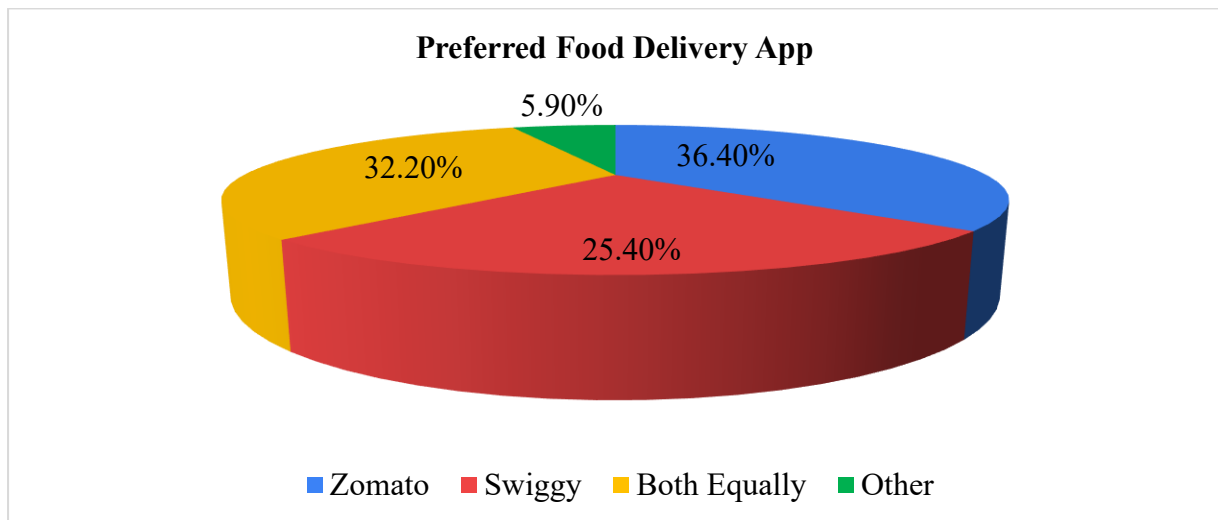
Data Interpretation :

The higher share of respondents above ₹80,000 suggests strong purchasing power and greater adoption of food delivery services. Balanced middle-income representation indicates widespread usage, while lower-income groups reflect growing accessibility and acceptance across diverse economic segments.

1. Which food delivery app do you use most frequently?

Data Table 4 : 118 Responses

Sr. No	App Used	Respondents	Percentage
1	Zomato	43	36.4%
2	Swiggy	30	25.4%
3	Both Equally	38	32.2%
4	Other	7	5.9%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The data shows that Zomato has the highest usage at 36.4%, followed by “Both Equally” at 32.2%, indicating multi-app behavior. Swiggy holds 25.4%, while other apps are minimal at 5.9%. The distribution suggests moderate competition, with no single platform dominating completely.

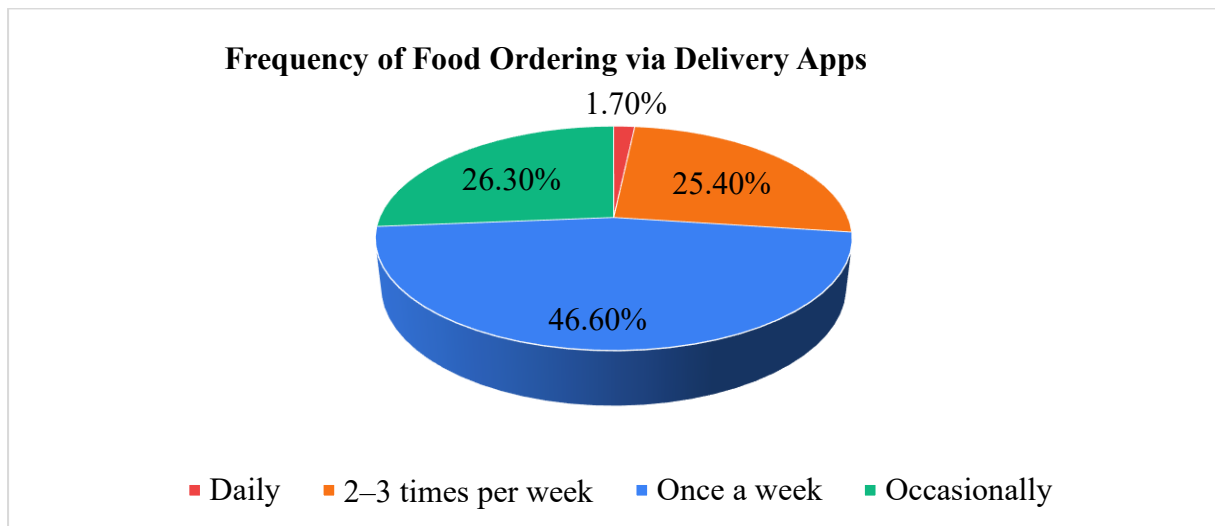
Data Interpretation :

Users do not strongly commit to a single platform, as a large share uses both Zomato and Swiggy. This indicates low switching costs and high price sensitivity. To increase loyalty, platforms must focus on personalization, offers, and user experience rather than relying solely on brand preference.

2. How often do you order food using food delivery apps?

Data Table 5 : 118 Responses

Sr. No	Frequency	Respondents	Percentage
1	Daily	2	1.70%
2	2–3 times per week	30	25.40%
3	Once a week	55	46.60%
4	Occasionally	31	26.30%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The data shows that most respondents order food once a week (46.6%), followed by occasional users (26.3%) and those ordering 2–3 times weekly (25.4%). Daily usage is minimal, indicating moderate but consistent engagement with food delivery apps.

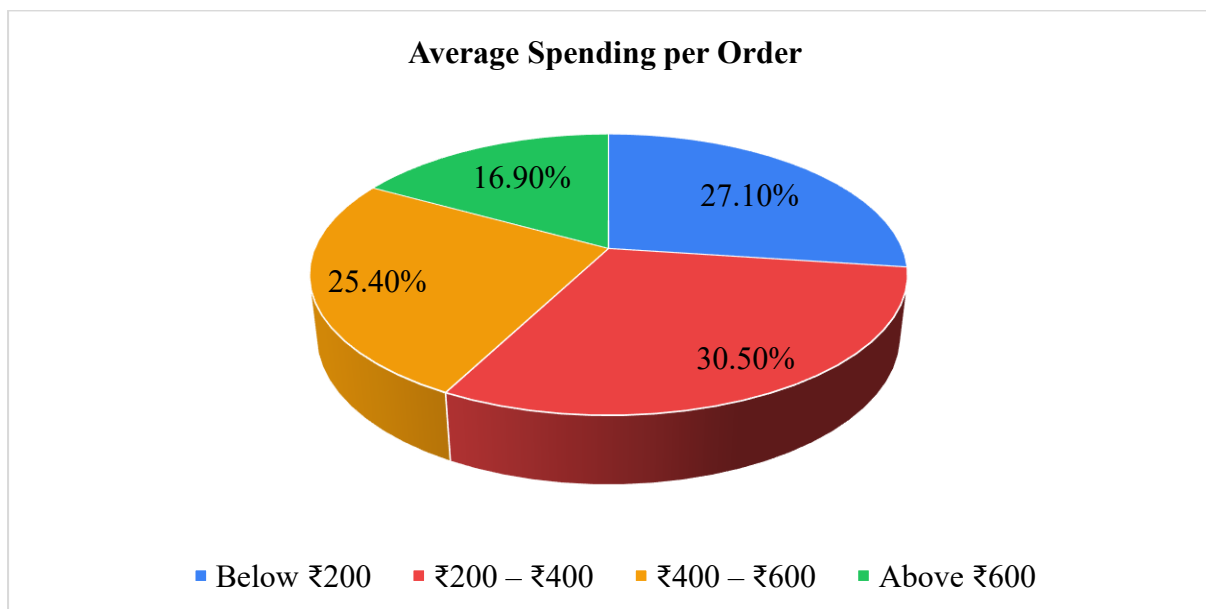
Data Interpretation :

The findings suggest that food delivery is primarily a weekly habit rather than a daily necessity. Users rely on these platforms for convenience and occasional indulgence, reflecting controlled spending behavior and situational usage rather than routine dependency.

3. What is your average spending per order?

Data Table 6 : 118 Responses

Sr. No	Spending Range	Respondents	Percentage
1	Below ₹200	32	27.1%
2	₹200 – ₹400	36	30.5%
3	₹400 – ₹600	30	25.4%
4	Above ₹600	20	16.9%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The majority of respondents (30.5%) spend between ₹200–₹400 per order, followed by 27.1% spending below ₹200. Mid-range spending ₹400–₹600 accounts for 25.4%, while only 16.9% of users spend above ₹600.

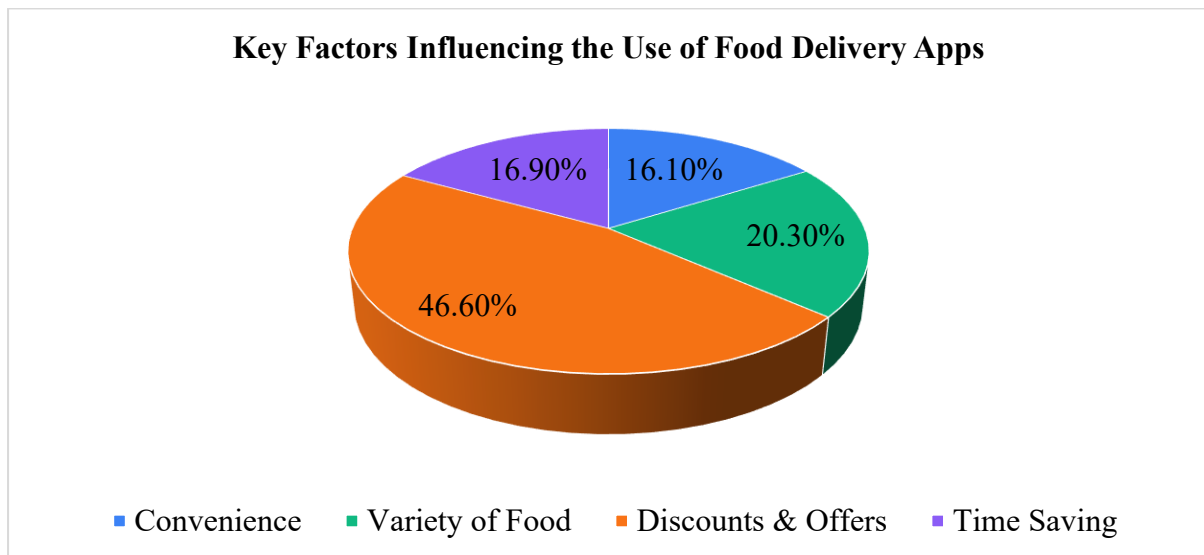
Data Interpretation :

Consumers mostly prefer moderate spending, indicating price sensitivity and value-conscious behavior. High spending is limited, suggesting users are influenced by affordability, discounts, and budget constraints rather than premium or luxury food ordering habits.

4. What is the main reason for using food delivery apps?

Data Table 7 : 118 Responses

Sr. No	Reason	Respondents	Percentage
1	Convenience	19	16.1%
2	Variety of Food	24	20.3%
3	Discounts & Offers	55	46.6%
4	Time Saving	20	16.9%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

Discounts and offers account for the largest share at 46.6%, making them the dominant reason for using food delivery apps. Variety of food contributes 20.3%, while convenience and time saving show similar but lower contributions at 16.1% and 16.9%.

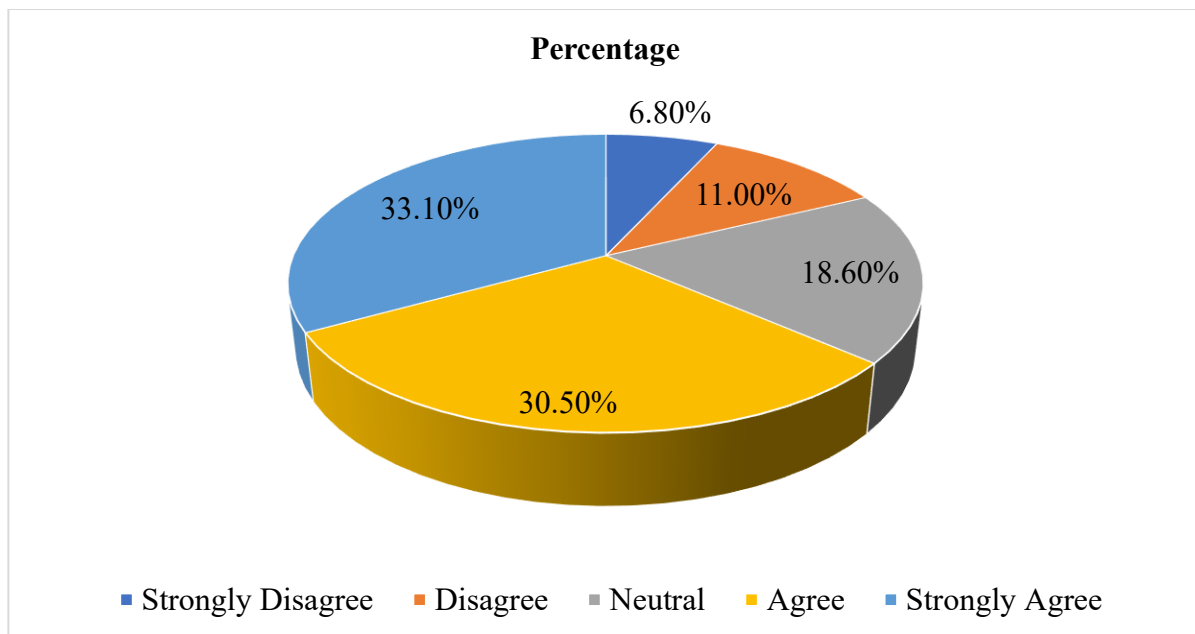
Data Interpretation :

The findings indicate that users are strongly influenced by promotional pricing rather than just convenience. Although variety and time efficiency matter, financial incentives significantly drive user engagement, highlighting the importance of discounts in attracting and retaining customers.

5. Food delivery apps recommend restaurants based on my preferences.

Data Table 8 : 118 Responses

Sr. No	Response Category	Respondents	Percentage
1	Strongly Disagree	8	6.8%
2	Disagree	13	11.0%
3	Neutral	22	18.6%
4	Agree	36	30.5%
5	Strongly Agree	39	33.1%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

Most respondents express positive sentiment, with 33.1% strongly agreeing and 30.5% agreeing that apps recommend based on preferences. Neutral responses account for 18.6%, while disagreement levels remain relatively low at 17.8% combined.

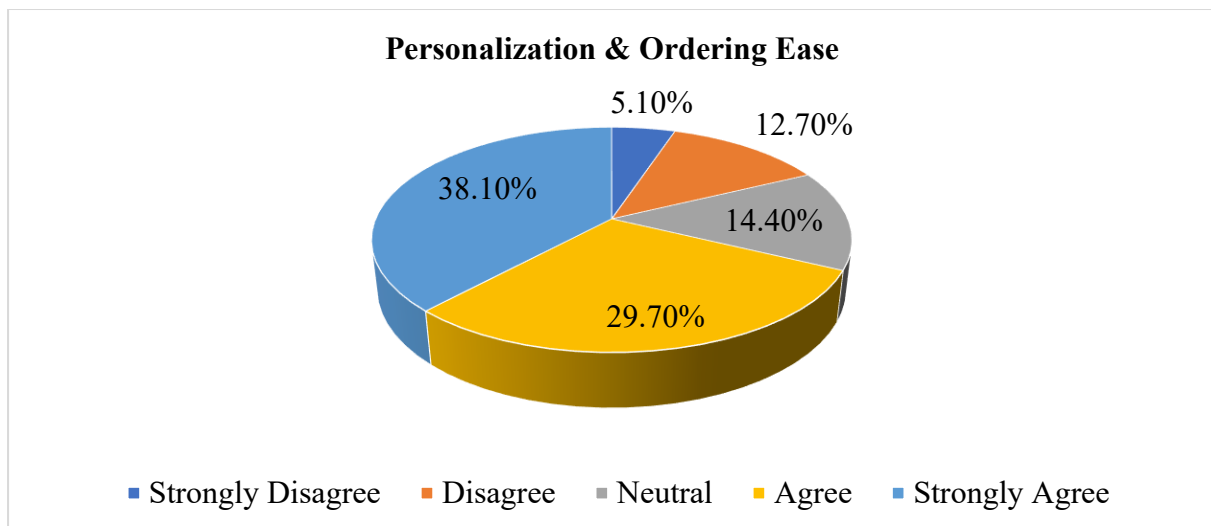
Data Interpretation :

The results indicate that recommendation systems are largely effective and trusted by users. A strong positive inclination suggests personalization enhances user experience, though some neutral and negative responses highlight scope for improving accuracy and relevance.

6. Personalized recommendations make ordering easier for me.

Data Table 9 : 118 Responses

Sr. No	Response Category	Respondents	Percentage
1	Strongly Disagree	6	5.1%
2	Disagree	15	12.7%
3	Neutral	17	14.4%
4	Agree	35	29.7%
5	Strongly Agree	45	38.1%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A strong majority of respondents show positive sentiment, with 38.1% strongly agreeing and 29.7% agreeing that personalized recommendations simplify ordering. Neutral responses are moderate, while disagreement remains comparatively low, indicating overall favorable perception.

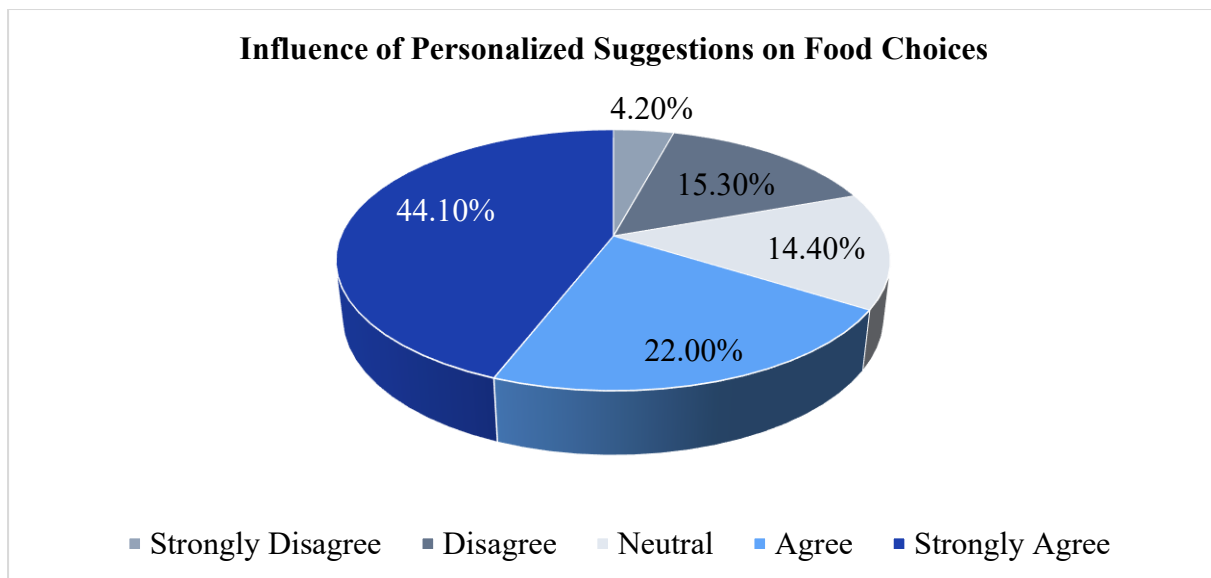
Data Interpretation :

The results suggest personalization significantly enhances user experience and decision-making. Most users rely on recommendations for convenience, though a small segment remains unconvinced, highlighting opportunities to refine recommendation accuracy and relevance further.

7. “Recommended for you” suggestions influence my food choices.

Data Table 10 : 118 Responses

Sr. No	Response Category	Respondents	Percentage
1	Strongly Disagree	5	4.2%
2	Disagree	18	15.3%
3	Neutral	17	14.4%
4	Agree	26	22.0%
5	Strongly Agree	52	44.1%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A significant proportion of respondents (44.1%) strongly agree that “recommended for you” suggestions influence their food choices, followed by 22% agreeing. Neutral and negative responses are comparatively lower, indicating a strong overall positive inclination toward recommendation influence.

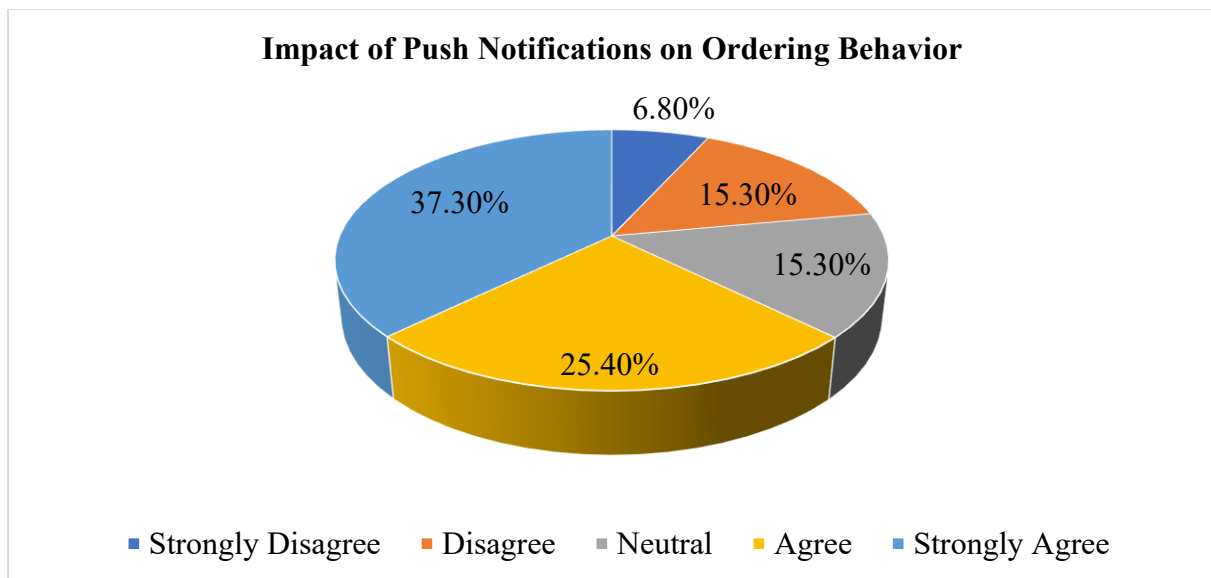
Data Interpretation :

The findings highlight that personalized suggestions play a major role in shaping customer decisions. Users are highly influenced by algorithm-driven recommendations, suggesting that effective personalization strategies can significantly impact ordering behavior and increase engagement on food delivery platforms.

8. Push notifications about offers encourage me to place orders.

Data Table 11 : 118 Responses

Sr. No	Response Category	Respondents	Percentage
1	Strongly Disagree	8	6.8%
2	Disagree	18	15.3%
3	Neutral	18	15.3%
4	Agree	30	25.4%
5	Strongly Agree	44	37.3%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A strong positive response is observed, with 37.3% strongly agreeing and 25.4% agreeing that push notifications influence ordering behavior. Neutral and negative responses are lower, though still notable, indicating varied sensitivity to promotional alerts.

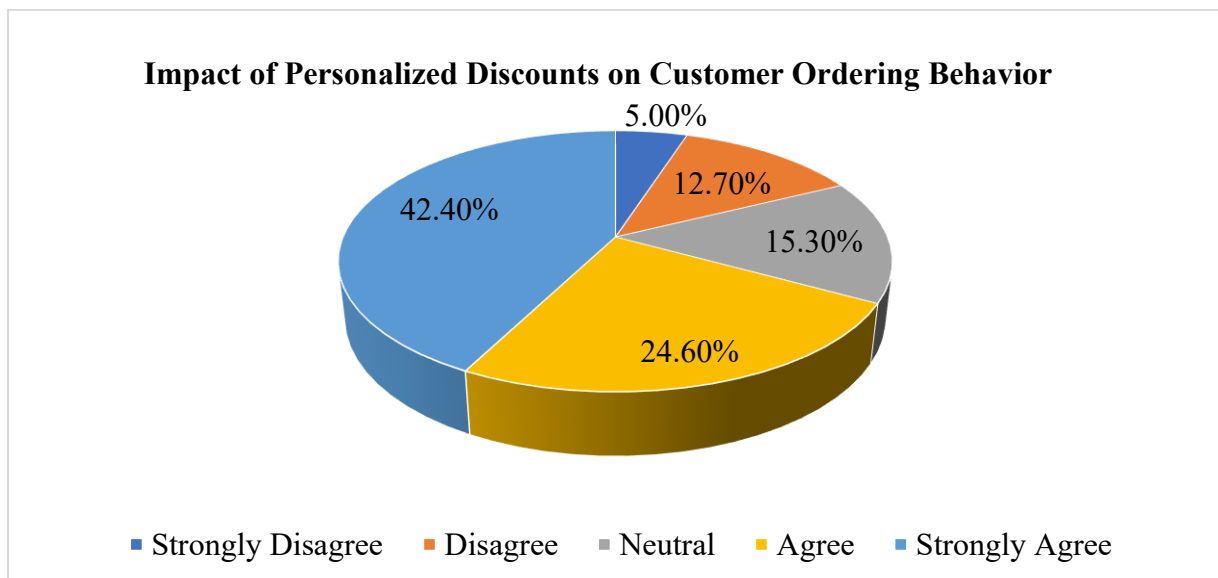
Data Interpretation :

Push notifications act as effective behavioral triggers, encouraging users to place orders through timely offers. However, moderate neutral and negative responses suggest that excessive or irrelevant notifications may reduce effectiveness, emphasizing the need for targeted and personalized messaging.

9. Personalized discounts increase my likelihood of ordering food

Data Table 12 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	6	5.0%
2	Disagree	15	12.7%
3	Neutral	18	15.3%
4	Agree	29	24.6%
5	Strongly Agree	50	42.4%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The majority of respondents (42.4%) strongly agree that personalized discounts increase ordering likelihood, followed by 24.6% agreeing. Only a small portion shows disagreement, indicating a strong positive inclination toward discount-driven purchasing behavior.

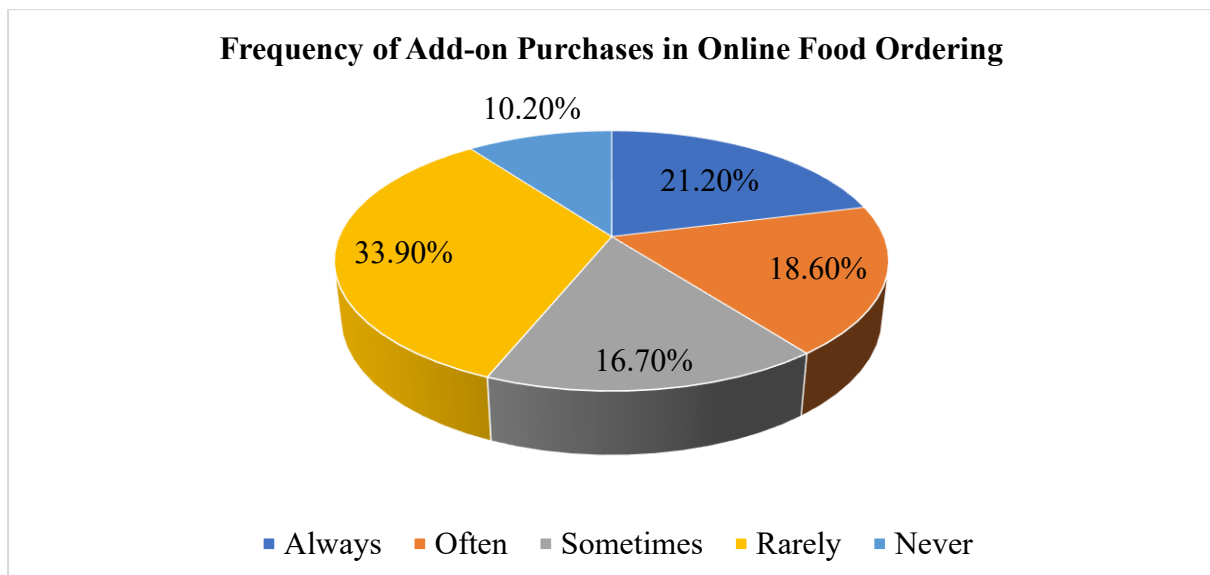
Data Interpretation :

Personalized discounts act as a powerful motivator, significantly influencing customer decisions. High agreement levels suggest that targeted offers enhance engagement and conversion, making discount personalization a key strategy for increasing order frequency in food delivery platforms.

10. How often do you purchase add-on items (desserts, beverages, side dishes) when ordering food online?

Data Table 13 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Always	25	21.2%
2	Often	22	18.6%
3	Sometimes	20	16.7%
4	Rarely	40	33.9%
5	Never	12	10.2%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

The largest segment of respondents (33.9%) rarely purchase add-on items, while 21.2% always and 18.6% often do. A moderate portion purchases occasionally, indicating mixed behavior with a tendency toward lower add-on purchasing frequency overall.

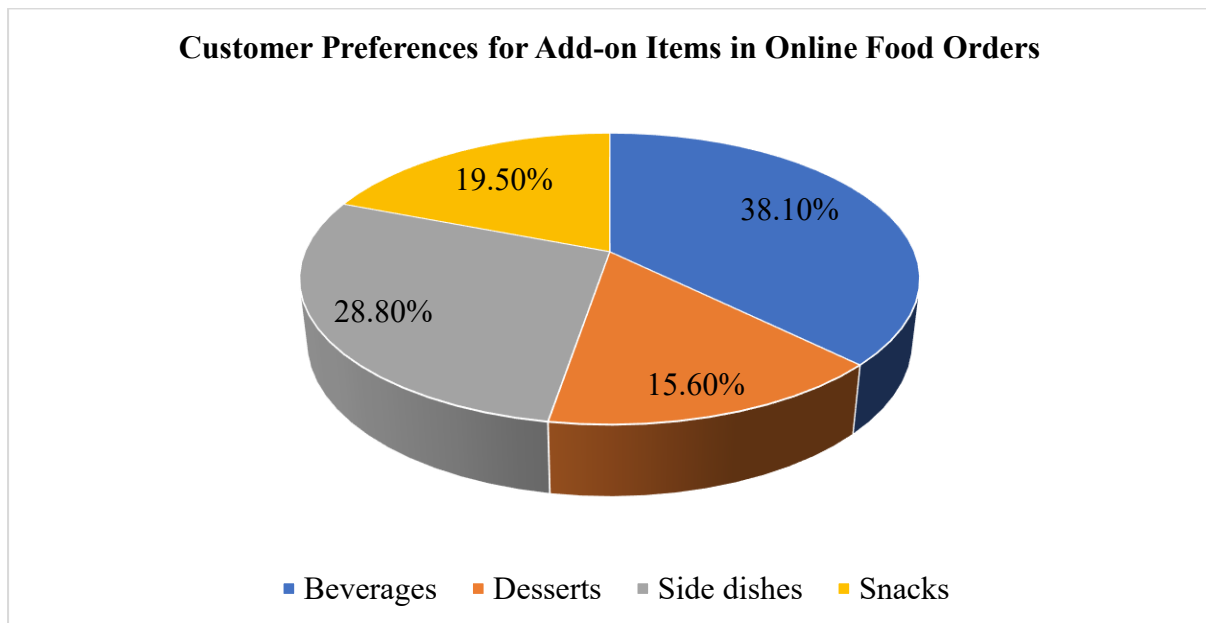
Data Interpretation :

Consumers show limited inclination toward add-on purchases, with many opting out or buying infrequently. This suggests a need for stronger incentives, bundling strategies, or personalized recommendations to increase add-on conversions and enhance overall order value.

11. Which add-on items do you usually purchase?

Data Table 14 : 118 Responses

Sr. No	Add-on Item	Respondents	Percentage (%)
1	Beverages	45	38.1%
2	Desserts	18	15.6%
3	Side dishes	34	28.8%
4	Snacks	23	19.5%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

Beverages dominate add-on purchases at 38.1%, followed by side dishes at 28.8%. Snacks account for 19.5%, while desserts are least preferred at 15.6%, indicating a stronger inclination toward complementary and practical add-on items.

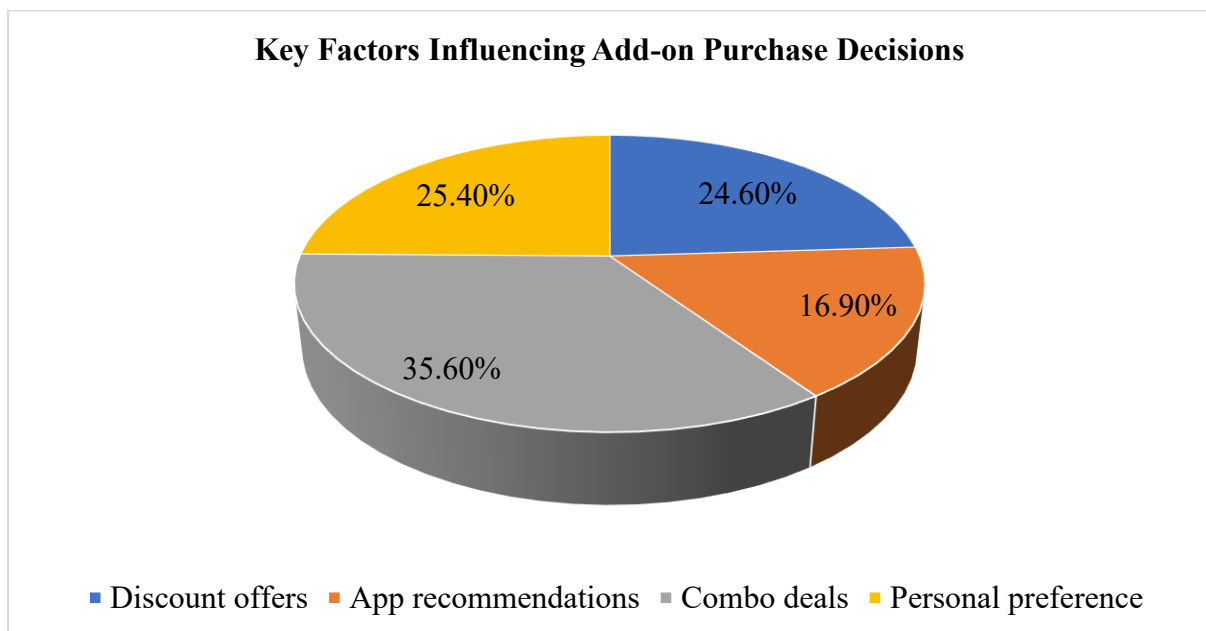
Data Interpretation :

Customers prefer functional add-ons like beverages and side dishes that enhance the main meal. Lower dessert preference suggests indulgence is secondary, highlighting opportunities for combo deals or targeted promotions to increase dessert and snack purchases.

12. What motivates you to buy add-on items?

Data Table 15 : 118 Responses

Sr. No	Factor	Respondents	Percentage (%)
1	Discount offers	29	24.6%
2	App recommendations	20	16.9%
3	Combo deals	42	35.6%
4	Personal preference	30	25.4%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

Combo deals emerge as the leading motivator at 35.6%, followed by personal preference (25.4%) and discount offers (24.6%). App recommendations contribute the least at 16.9%, indicating stronger influence of pricing strategies over algorithm-driven suggestions.

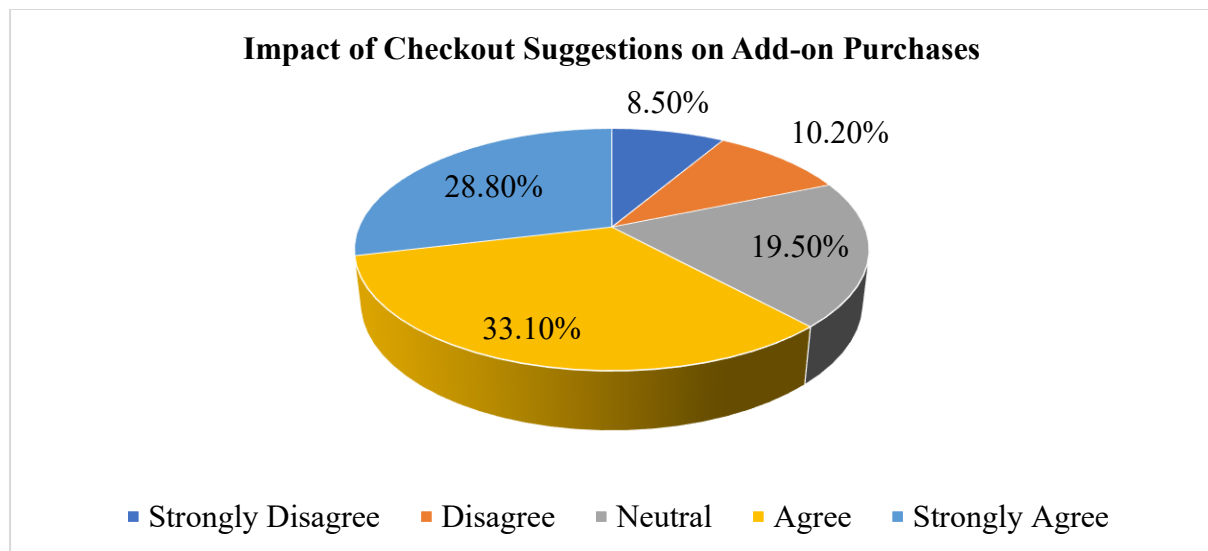
Data Interpretation :

Customers are primarily driven by value-based incentives like combo deals and discounts rather than app recommendations. This highlights the importance of bundling and pricing strategies to boost add-on sales, while recommendation systems may require better personalization to improve effectiveness.

13. I often purchase add-on items suggested during checkout.

Data Table 16 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	10	8.5%
2	Disagree	12	10.2%
3	Neutral	23	19.5%
4	Agree	39	33.1%
5	Strongly Agree	34	28.8%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A majority of respondents (33.1% agree and 28.8% strongly agree) purchase add-on items suggested during checkout. Neutral responses are moderate, while a smaller portion disagrees, indicating that checkout suggestions significantly influence customer purchase behavior.

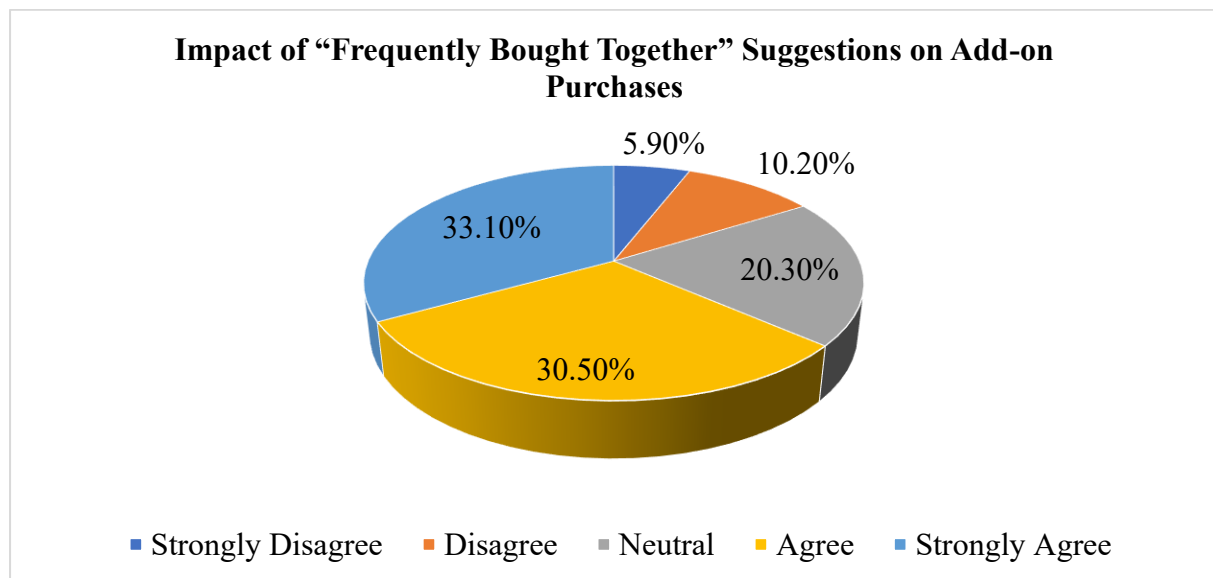
Data Interpretation :

Checkout-stage recommendations effectively drive impulse add-on purchases, leveraging customer readiness to buy. High agreement suggests this is a critical conversion point, where well-timed and relevant suggestions can significantly enhance order value and overall sales performance.

14. “Frequently bought together” suggestions influence my decision to buy add-ons.

Data Table 17 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	7	5.9%
2	Disagree	12	10.2%
3	Neutral	24	20.3%
4	Agree	36	30.5%
5	Strongly Agree	39	33.1%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A strong majority of respondents (33.1% strongly agree and 30.5% agree) are influenced by “frequently bought together” suggestions. Neutral responses are moderate, while a small percentage disagrees, indicating a clear positive impact of bundled recommendations.

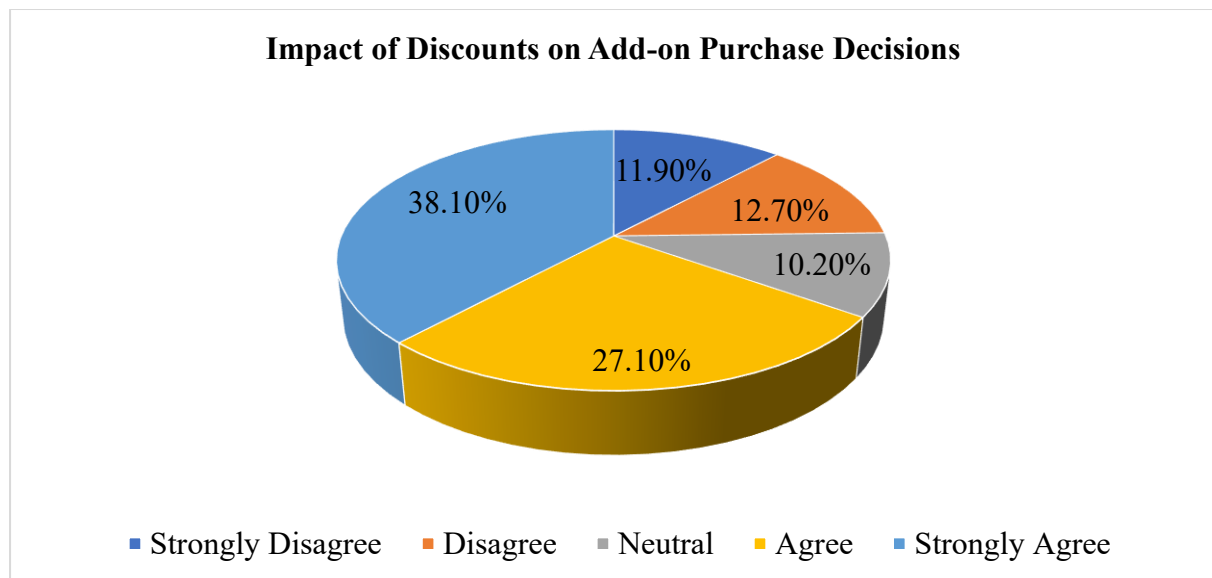
Data Interpretation :

“Frequently bought together” suggestions effectively drive add-on purchases by simplifying decision-making and highlighting complementary items. High agreement indicates that such recommendation strategies enhance perceived value and convenience, increasing the likelihood of additional purchases during ordering.

15. Discounts increase my likelihood of purchasing add-on items

Data Table 18 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	14	11.9%
2	Disagree	15	12.7%
3	Neutral	12	10.2%
4	Agree	32	27.1%
5	Strongly Agree	45	38.1%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A majority of respondents (38.1% strongly agree and 27.1% agree) indicate that discounts increase their likelihood of purchasing add-on items. Neutral responses are lower, while a moderate portion disagrees, highlighting the strong influence of pricing incentives.

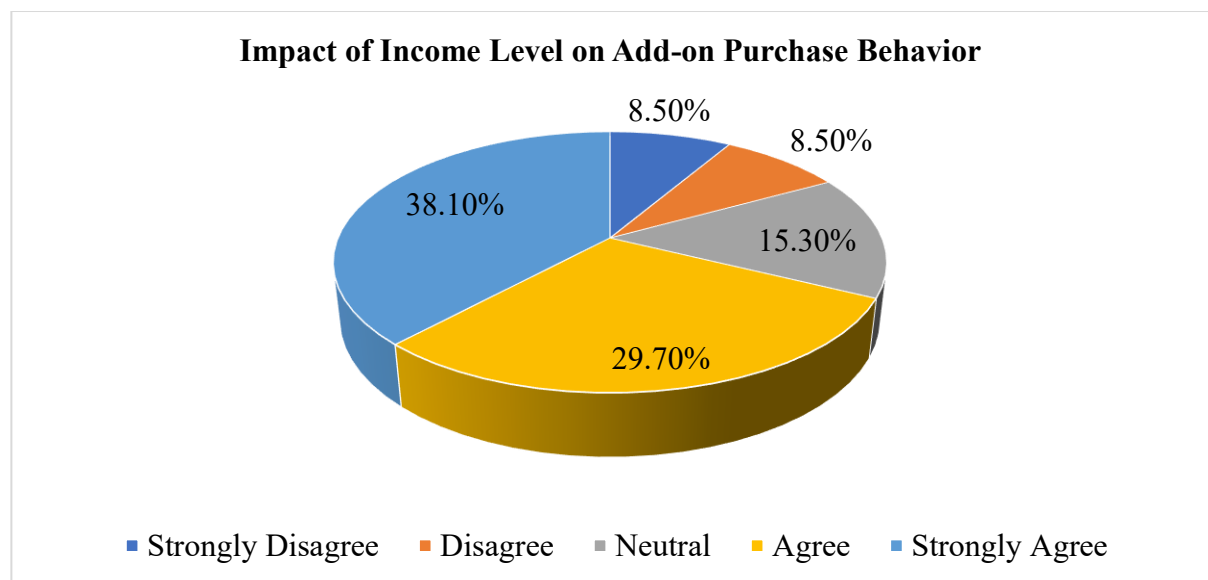
Data Interpretation :

Discounts serve as a key motivator for add-on purchases by enhancing perceived value and reducing price sensitivity. High agreement suggests that strategic discounting can effectively boost add-on sales and increase overall order value in online food delivery platforms.

16. My income level affects whether I purchase add-on items

Data Table 19 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	10	8.5%
2	Disagree	10	8.5%
3	Neutral	18	15.3%
4	Agree	35	29.7%
5	Strongly Agree	45	38.1%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A majority of respondents (38.1% strongly agree and 29.7% agree) believe income level affects add-on purchases. Neutral responses are moderate, while a smaller percentage disagrees, indicating income plays a significant role in influencing purchasing behavior.

Data Interpretation :

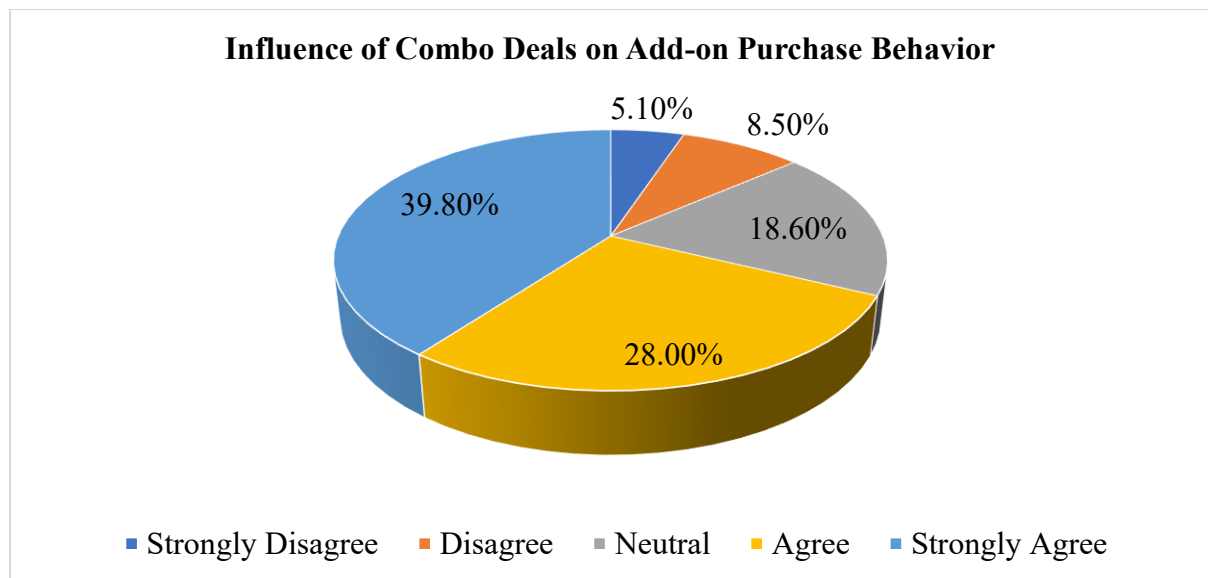
The results suggest that purchasing add-on items is influenced by consumers' financial capacity and spending flexibility. Higher agreement indicates that affordability impacts decision-making, making income an important factor in determining willingness to spend on additional items.

lower-income groups may limit discretionary purchases, while higher-income consumers show greater flexibility in adding extras.

17. I buy add-on items when they are offered as combo deals.

Data Table 20 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	6	5.1%
2	Disagree	10	8.5%
3	Neutral	22	18.6%
4	Agree	33	28.0%
5	Strongly Agree	47	39.8%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A large proportion of respondents (39.8% strongly agree and 28.0% agree) purchase add-on items when offered as combo deals. Neutral responses are moderate, while a small percentage disagrees, indicating strong effectiveness of bundled pricing strategies.

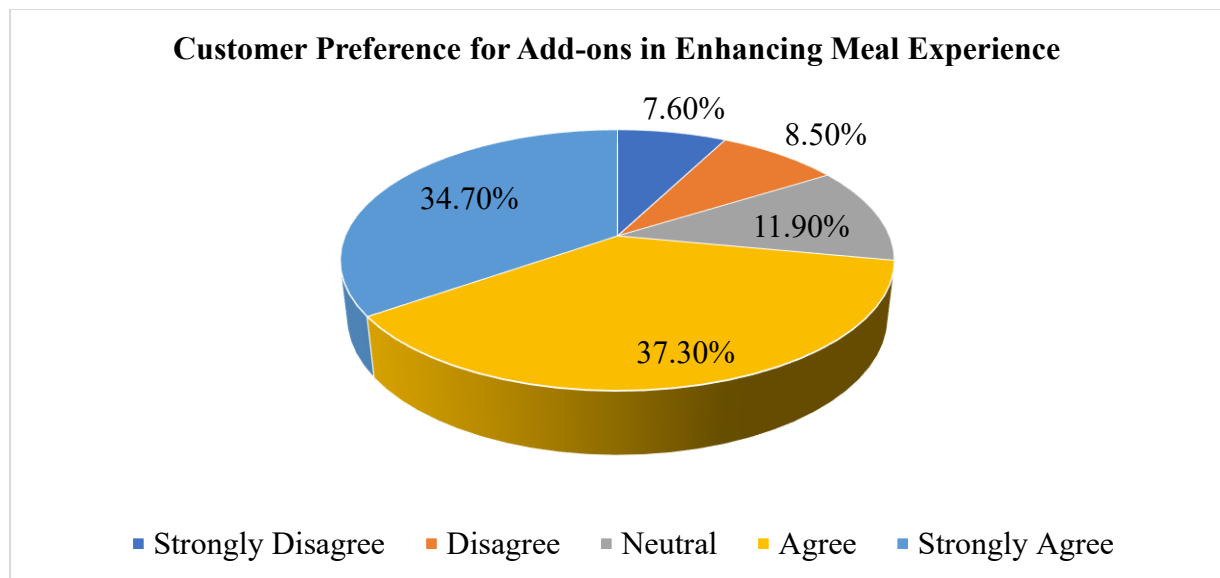
Data Interpretation :

Combo deals significantly influence customer purchasing behavior by offering perceived value and convenience. High agreement suggests that bundling strategies encourage customers to buy additional items, increasing both order size and satisfaction levels during the purchasing process. Combo deals simplify decision-making and create a perception of savings, further enhancing customer engagement and purchase likelihood.

18. I prefer ordering add-ons to complete my meal experience

Data Table 21 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	9	7.6%
2	Disagree	10	8.5%
3	Neutral	14	11.9%
4	Agree	44	37.3%
5	Strongly Agree	41	34.7%
	Total	118	100%



Data Analysis and Interpretation

Data Analysis :

A significant majority of respondents (37.3% agree and 34.7% strongly agree) prefer ordering add-ons to complete their meal experience. Neutral responses are limited, while a small portion disagrees, indicating a strong positive inclination toward add-ons.

Data Interpretation :

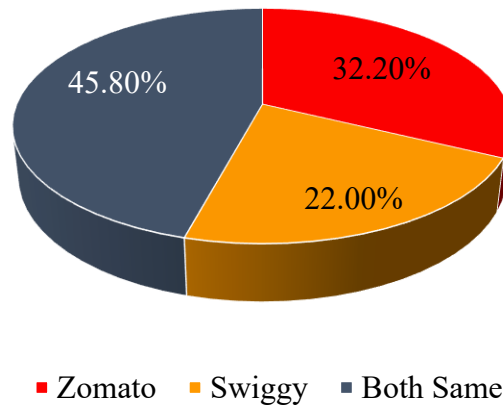
Customers perceive add-ons as an essential part of enhancing their overall meal experience. High agreement indicates that add-ons are not just optional but contribute to satisfaction, encouraging platforms to strategically promote complementary items alongside main orders. This behavior reflects a preference for complete, value-rich meals rather than standalone items.

19. Which platform provides better personalized recommendations?

Data Table 22 : 118 Responses

Sr. No	Platform	Respondents	Percentage (%)
1	Zomato	38	32.2%
2	Swiggy	26	22.0%
3	Both Same	54	45.8%
	Total	118	100%

Comparison of Personalized Recommendation Effectiveness Across Platforms



Data Analysis and Interpretation

Data Analysis :

The largest share of respondents (45.8%) believe both platforms provide similar personalized recommendations. Zomato leads individually at 32.2%, while Swiggy trails at 22%, indicating limited differentiation in perceived recommendation effectiveness. This distribution suggests that users experience comparable recommendation quality across platforms, reducing the competitive gap in personalization capabilities.

Data Interpretation :

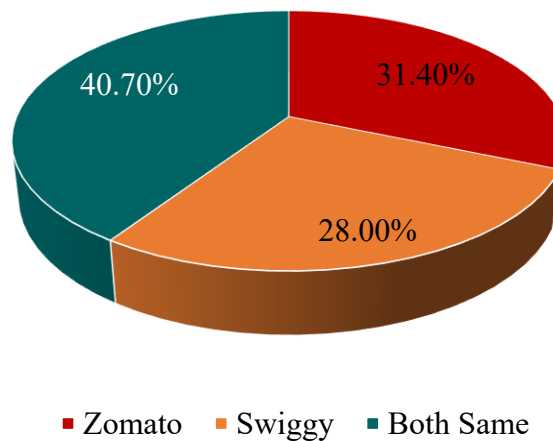
The findings suggest that users perceive minimal difference between platforms in terms of personalization quality. While Zomato has a slight edge, the dominance of “both same” indicates that recommendation systems across platforms are equally competitive and comparable. Additionally, this parity highlights an opportunity for platforms to innovate, improve algorithm accuracy, and create more distinct, personalized user experiences.

20. Which platform encourages more add-on purchases?

Data Table 23 : 118 Responses

Sr. No	Platform	Respondents	Percentage (%)
1	Zomato	37	31.4%
2	Swiggy	33	28.0%
3	Both Same	48	40.7%
	Total	118	100%

Platform Comparison in Encouraging Add-on Purchases



Data Analysis and Interpretation

Data Analysis :

The largest group of respondents (40.7%) believes both platforms equally encourage add-on purchases, indicating no strong differentiation. Zomato leads slightly at 31.4%, followed by Swiggy at 28%. This distribution suggests that while both platforms perform similarly, Zomato has a marginal advantage in influencing add-on buying behavior.

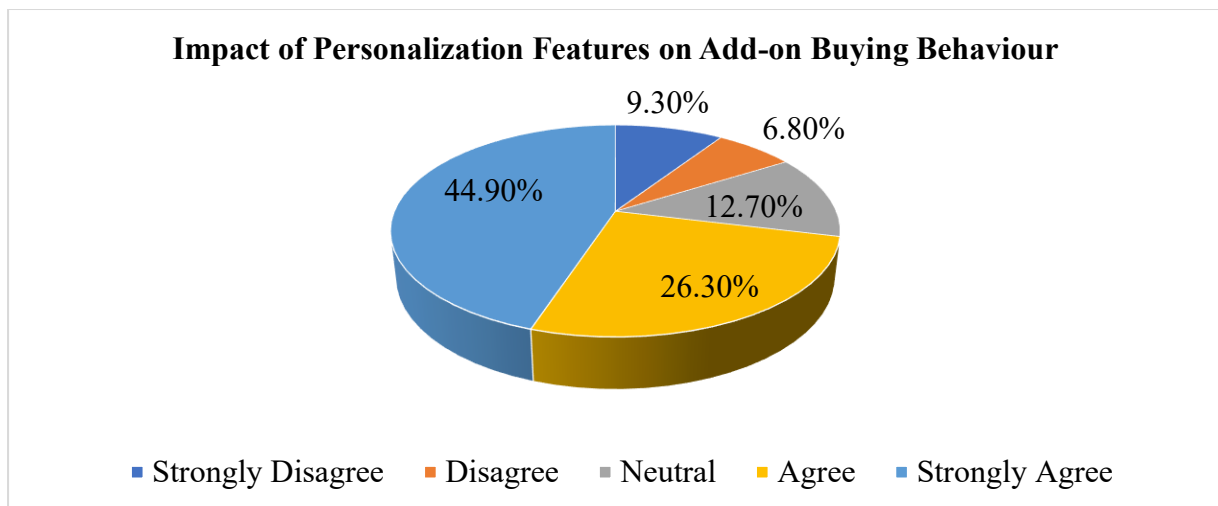
Data Interpretation :

The findings indicate that both platforms are perceived to have similar effectiveness in encouraging add-on purchases, reflecting comparable marketing and recommendation strategies. Although Zomato shows a slight edge, the dominance of “both same” responses highlights limited competitive distinction. This suggests opportunities for innovation in personalized offers, bundling, and user engagement strategies.

21. Overall, personalization features influence my add-on buying behaviour.

Data Table 24 : 118 Responses

Sr. No	Response	Respondents	Percentage (%)
1	Strongly Disagree	11	9.3%
2	Disagree	8	6.8%
3	Neutral	15	12.7%
4	Agree	31	26.3%
5	Strongly Agree	53	44.9%
	Total	118	100%



Data Analysis and Interpretation

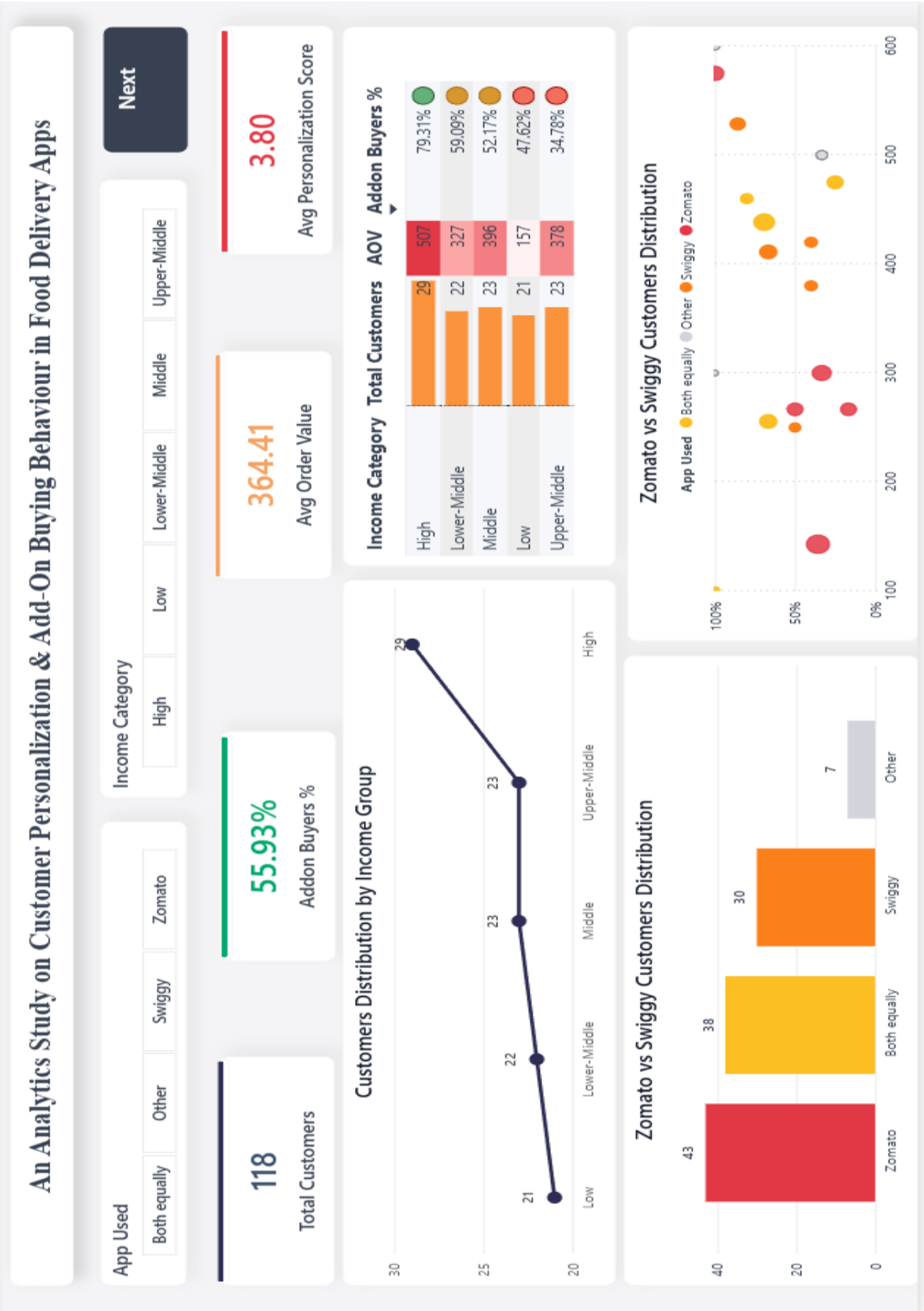
Data Analysis :

A clear majority of respondents (44.9% strongly agree and 26.3% agree) indicate that personalization features influence their add-on buying behaviour. Neutral responses remain moderate at 12.7%, while disagreement is relatively low. This distribution suggests a strong positive inclination toward personalization, although a non-trivial segment remains unconvinced or indifferent.

Data Interpretation :

Personalization appears to significantly shape customer perception of add-on purchasing decisions, but this does not guarantee actual purchase behavior. While high agreement suggests strong influence, it likely works best when combined with pricing incentives and timing. Platforms relying solely on personalization without value-driven triggers may see limited real conversion impact.

Primary Data Dashboard



An Analytics Study on Customer Personalization & Add-On Buying Behaviour in Food Delivery Apps

App Used

Both equally

Other

Swiggy

Zomato

Income Category

High

Low

Lower-Middle

Middle

Upper-Middle

Back

118

Total Customers

55.93%

Addon Buyers %

364.41

Avg Order Value

3.80

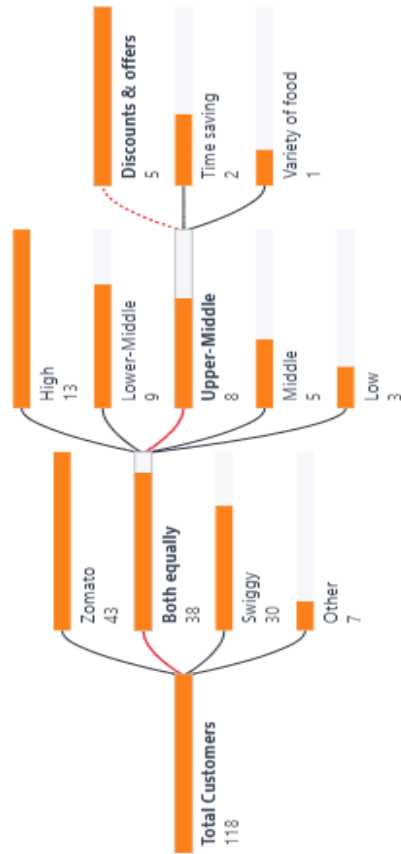
Avg Personalization Score

Customer Breakdown Analysis

App Used ☒ Both equally ☒ Income Category ☒ Primary Usage ☒

Both equally

Upper-Middle



Secondary Data Dashboard



RESULTS

Hypothesis Testing

1 : To compare customer add-on preferences between Zomato and Swiggy.

Add - On	Swiggy	Zomato
Beverages	11.92	17.08
Desserts	3.70	5.30
Side dishes	9.04	12.96
Snacks	5.34	7.66

$$0.1438 > 0.05$$

The Chi-square test was conducted to examine whether there is a significant difference in add-on preferences between Zomato and Swiggy users. The calculated p-value (0.1438) is greater than the significance level of 0.05. Therefore, the null hypothesis is accepted. This indicates that there is no statistically significant difference in add-on preferences between users of Zomato and Swiggy.

2 : To evaluate personalization features of food delivery apps.

Personalization	No	Yes
No	9.51	24.49
Yes	23.49	60.51

$$0.0000000488 < 0.05$$

The Chi-square test was conducted to examine the influence of personalization features on add-on buying behaviour. The calculated p-value is less than 0.05, indicating a statistically significant relationship between personalization and add-on purchases. Therefore, the null hypothesis is rejected, and it can be concluded that personalization features significantly influence add-on buying decisions.

3: To assess determinants of customers' add-on spending patterns.

Discount	NO	YES
NO	8.39	21.61
YES	24.61	63.39

$$\begin{aligned} \text{p-value} &= 6.29662\text{E-}10 \\ &\approx 0.0000000006297 \end{aligned}$$

The Chi-square test was conducted to assess the effect of determinants such as discounts on add-on buying behaviour. The calculated p-value is less than 0.05, indicating a statistically significant relationship between pricing factors and add-on purchases. Therefore, the null hypothesis is rejected. It can be concluded that determinants such as discounts and combo offers significantly influence customers' add-on spending patterns.

FINDINGS

(Findings for Primary Data)

1. Gender

Male respondents (61.9%) slightly dominate the sample, while females account for 38.1%. This indicates that food delivery apps are widely used across both genders, though slightly more among males.

2. Occupation

Working professionals form the largest group (35.6%), followed by business and students. This suggests higher dependence on food delivery services due to busy schedules and convenience needs.

3. Income

The highest proportion of respondents (24.6%) fall in the above ₹80,000 category. This reflects strong purchasing power and higher adoption of food delivery platforms among higher-income users.

4. App Usage

Zomato leads with 36.4%, but 32.2% of users use both platforms equally. This indicates low platform loyalty and high switching behavior based on offers and experience.

5. Ordering Frequency

A majority of users (46.6%) order once a week, followed by occasional users (26.3%). This shows that food delivery is a habitual but not daily activity for most consumers.

6. Spending per Order

Most users (30.5%) spend between ₹200–₹400 per order. This reflects price-sensitive behavior and a preference for moderate spending.

7. Reason for Using Apps

Discounts and offers dominate with 46.6%, making them the primary driver of app usage. This highlights strong price sensitivity and dependence on promotional incentives.

8. Recommendation Accuracy

Around 63.6% of users agree that apps recommend based on their preferences. This indicates that personalization systems are generally effective but still have room for improvement.

9. Ease of Ordering

Approximately 67.8% of respondents agree that personalized recommendations make ordering easier.

This shows that personalization enhances convenience and simplifies decision-making.

10. Influence of Suggestions

A significant 66.1% of users agree that “recommended for you” suggestions influence their choices.

This confirms that algorithm-driven recommendations play an important role in shaping decisions.

11. Push Notifications

Nearly 62.7% of respondents agree that notifications influence their ordering behavior. However, effectiveness depends on relevance and timing of these notifications.

12. Personalized Discounts

About 67% of users agree that personalized discounts increase their likelihood of ordering. This highlights discounts as a strong conversion driver when combined with personalization.

13. Add-on Purchase Frequency

A large portion of users (33.9%) rarely purchase add-ons, while only 21.2% always do. This indicates that add-on buying is not habitual and requires active triggers.

14. Add-on Preferences

Beverages (38.1%) and side dishes (28.8%) are the most preferred add-ons. Customers prefer practical, meal-complementing items over indulgent options like desserts.

15. Add-on Motivation

Combo deals lead with 35.6%, followed by discounts and personal preference. This shows that value-based bundling is more effective than standalone recommendations.

16. Checkout Suggestions

Around 61.9% of users agree that checkout suggestions influence their purchases. This highlights the checkout stage as a key point for driving impulse buying.

17. “Frequently Bought Together”

Approximately 63.6% of respondents agree that bundled suggestions influence decisions. These features reduce decision effort and increase the likelihood of add-on purchases.

18. Discounts on Add-ons

About 65.2% of users agree that discounts increase add-on purchases. This reinforces the importance of pricing strategies in influencing buying behavior.

19. Income Effect

Nearly 67.8% of respondents agree that income affects add-on buying decisions. Higher-income users show greater willingness to spend on additional items.

20. Combo Deals Impact

Around 67.8% of users agree that combo deals encourage add-on purchases. This confirms bundling as one of the most effective strategies for increasing order value.

21. Add-ons as Experience

A strong 72% of users agree that add-ons enhance their meal experience. This suggests add-ons are perceived as part of a complete meal rather than optional extras.

22. Personalization Comparison

About 45.8% of users believe both platforms provide similar personalization. This indicates low differentiation between Zomato and Swiggy in recommendation quality.

23. Add-on Influence by Platform

Around 40.7% of respondents feel both platforms equally influence add-on purchases. This shows similar performance in driving add-on behavior across platforms.

24. Overall Personalization Impact

A strong 71.2% of users agree that personalization influences their behavior. However, its effectiveness increases significantly when combined with pricing strategies.

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(Findings for Secondary Data)

1. Platform Usage vs Performance

Zomato leads usage (36.4%), but Swiggy shows higher add-on rate (77.17% vs 59.23%), indicating user preference does not directly translate into higher revenue performance.

2. Ordering Frequency Impact

Most users order weekly (46.6%), while dashboard shows higher frequency users generate higher AOV (₹258 → ₹377), highlighting frequency as a key revenue driver.

3. Spending Behavior

Majority spending lies between ₹200–₹400 (30.5%), supported by dashboard AOV ₹314, confirming moderate and price-sensitive customer behavior.

4. Discounts as Primary Driver

Discounts influence 46.6% users, and add-on rate increases from 54.83% to 73.24% with offers, proving pricing incentives are the strongest purchase trigger.

5. Personalization Role

Around 70% users respond positively to personalization, but its impact is secondary and more effective when combined with pricing and promotional strategies.

6. Add-on Purchase Behavior

33.9% users rarely buy add-ons, while dashboard shows 68.17% add-on rate, indicating a gap between perceived and actual purchase behavior.

7. Add-on Preferences

Beverages (38.1%) and side dishes (28.8%) dominate, aligning with dashboard data, showing preference for practical add-ons over indulgent items.

8. Effect of Combo Deals

Combo deals influence 35.6% users, with strong agreement (39.8%), and higher add-on rates in premium segments (77.75%), making bundling highly effective.

9. Income Influence

67.8% users agree income affects purchases, supported by AOV rising from ₹276 (low) to ₹365 (high), confirming affordability impacts add-on buying.

10. Platform Differentiation

Most users perceive both platforms as similar (45.8% personalization, 40.7% add-ons), indicating low differentiation and a highly competitive, parity-driven market.

CONCLUSION

The study examines the impact of personalization on add-on buying behaviour using both primary survey data (118 respondents) and secondary dashboard data (30,000 customers). The findings from both sources consistently indicate that personalization plays a meaningful role in influencing customer decisions, but it is not the primary driver of add-on purchases.

Primary data reveals that a majority of respondents (over 70%) perceive personalization features such as recommendations, notifications, and suggestions as helpful in simplifying decision-making and enhancing the ordering experience. However, customers do not consistently translate this perception into actual add-on purchases, as a significant proportion (33.9%) report rarely purchasing add-ons.

Secondary data provides stronger behavioral evidence, showing a relatively high add-on purchase rate (68.17%) and increased engagement among users exposed to personalization features (72.46%). Despite this, the dashboard clearly demonstrates that pricing strategies have a stronger influence, with add-on purchase rates increasing significantly when discounts and offers are applied (from 54.83% to 73.24%).

Furthermore, both datasets highlight the importance of combo deals and bundled offers, which emerge as the most effective conversion drivers. Customers respond positively to value-based incentives that reduce decision effort and enhance perceived benefits. Additionally, income level plays a significant role, as higher-income groups generate higher average order values (₹365 compared to ₹276), indicating that affordability directly affects add-on purchasing behavior.

Platform comparison across both datasets shows minimal differentiation between Zomato and Swiggy, with a large proportion of users perceiving both platforms as similar in terms of personalization and add-on influence. However, secondary data suggests that performance differences exist, as Swiggy demonstrates higher add-on conversion rates despite similar user perception.

Overall, the study concludes that personalization acts as a supporting mechanism rather than a primary driver. While it enhances user experience and influences decision-making, actual conversion is driven by pricing strategies, combo deals, and timely recommendations at the checkout stage.

SUGGESTIONS

(Suggestion for Primary Data)

1. Shift from Generic Discounts to Smart, Targeted Offers

Blanket discounts waste margin. Use personalized discounts based on user behavior, cart value, and frequency to maximize conversion without over-discounting.

2. Prioritize Combo Deals as the Core Strategy

Stop treating combos as optional. Make bundled meals (main + beverage + side) the default upsell with clear savings to increase AOV and add-on adoption.

3. Optimize Checkout for Last-Moment Conversions

Most decisions happen here. Add one-click add-ons, “frequently bought together,” and urgency cues to convert users who otherwise skip add-ons.

4. Focus on High-Performing Add-ons Only

Don’t push everything. Promote beverages and sides (high demand) instead of low-performing items unless bundled or discounted.

5. Segment Users Based on Spending Power

Treat all users the same and you lose efficiency.
Offer:

- Premium bundles to high-income users
- Budget deals/discounts to price-sensitive users

(Suggestion for Secondary Data)

1. Improve Zomato's Add-on Conversion

Zomato should strengthen checkout recommendations and combo visibility to close the gap with Swiggy (59.23% vs 77.17%).

2. Focus on High-Income Segments

Target high-income users with premium bundles and higher-value combos since they generate the highest AOV (₹365).

3. Combine Personalization with Offers

Use personalized discounts and combo suggestions together, as add-on rates are higher with personalization + offers (73.24% vs 54.83%).

4. Increase Customer Order Frequency

Introduce loyalty programs, subscriptions, and reorder reminders to move users from occasional to frequent ordering, increasing AOV.

5. Promote High-Demand Add-ons

Highlight beverages and popular items in recommendations instead of low-performing add-ons to improve conversion rates.

DISCUSSION

1. Overview of Significant Findings

The study shows that while personalization features are positively perceived by users ($\approx 70\%$ agreement), they are not the primary drivers of add-on purchases. Instead, pricing factors such as discounts (46.6%) and combo deals (35.6%) have the strongest influence. Add-on purchases are not habitual (33.9% rarely purchase), but are significantly affected by checkout-stage suggestions and bundling strategies. Additionally, income level impacts spending behavior, with higher-income users showing greater willingness to purchase add-ons.

2. Findings in Light of Previous Research

The results align with existing research in consumer behavior and e-commerce, which suggests that price sensitivity and perceived value are key determinants of purchase decisions. Prior studies indicate that personalization enhances user experience but often acts as a supporting factor rather than a primary motivator, which is consistent with the findings of this study. Similarly, the effectiveness of bundling and promotional offers in increasing purchase likelihood has been widely documented, supporting the observed impact of combo deals.

3. Findings That Do Not Fully Support Expectations

Although personalization was expected to be a major driver of add-on purchases, the findings indicate that its impact is limited when used independently. Users acknowledge its usefulness, but actual purchase behavior is more strongly influenced by pricing incentives. Additionally, despite users perceiving personalization features positively, platform differentiation remains low, with many respondents indicating both platforms offer similar experiences.

4. Limitations of the Study

The study is based on a sample size of 118 respondents, which may not fully represent the broader population. The use of self-reported data (Likert scale responses) may introduce bias, as respondents' stated preferences may differ from actual behavior. Furthermore, the study focuses on selected variables and does not consider other factors such as geographic location, restaurant availability, or delivery time, which may also influence purchasing decisions.

5. Recommendations for Further Research

Future studies can expand the sample size and include diverse geographic regions to improve generalizability. Researchers may also use behavioral data (actual transaction data) instead of relying solely on survey responses. Additionally, further research can explore the role of real-time personalization, AI-driven recommendations, and user interface design in influencing add-on purchases.

6. Implications for Professional Practice

The findings suggest that food delivery platforms should focus on integrating personalization with pricing strategies rather than relying on personalization alone. Businesses can improve add-on sales by implementing targeted discounts, combo deals, and checkout-stage recommendations. Additionally, segmenting customers based on income and behavior can help design more effective marketing and upselling strategies, ultimately enhancing revenue and customer satisfaction.

Annexure

Questionnaire

Demographic Information

i). Gender

- ☐ Male
- ☐ Female
- ☐ Prefer not to say

ii). Occupation

- ☐ Student
- ☐ Working Professional
- ☐ Business
- ☐ Homemaker
- ☐ Other

iii). Monthly Income

- ☐ Below ₹20,000
- ☐ ₹20,000 – ₹40,000
- ☐ ₹40,000 – ₹60,000
- ☐ ₹60,000 – ₹80,000
- ☐ Above ₹80,000

1. Which food delivery app do you use most frequently?

- ☐ Zomato
- ☐ Swiggy
- ☐ Both equally
- ☐ Other

2. How often do you order food using delivery apps?

- ☐ Daily
- ☐ 2–3 times per week
- ☐ Once a week
- ☐ Occasionally

3. What is your average spending per order?

- ☐ Below ₹200
- ☐ ₹200 – ₹400
- ☐ ₹400 – ₹600
- ☐ Above ₹600

4. What is the main reason for using food delivery apps?

- ☐ Convenience
- ☐ Variety of food
- ☐ Discounts & offers
- ☐ Time saving

5. Food delivery apps recommend restaurants based on my preferences ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

6. Personalized recommendations make ordering easier for me ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

7. “Recommended for you” suggestions influence my food choices ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

8. Push notifications about offers encourage me to place orders ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

9. Personalized discounts increase my likelihood of ordering food ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

13. How often do you purchase add-on items (desserts, beverages, side dishes)?

- ☐ Always
- ☐ Often
- ☐ Sometimes
- ☐ Rarely
- ☐ Never

14. Which add-on items do you usually purchase?

- ☐ Beverages
- ☐ Desserts
- ☐ Side dishes
- ☐ Snacks
- ☐

15. What motivates you to buy add-on items?

- ☐ Discount offers
- ☐ App recommendations
- ☐ Combo deals
- ☐ Personal preference

16. I often purchase add-on items suggested during checkout ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

17. “Frequently bought together” suggestions influence my decision ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

18. Discounts increase my likelihood of purchasing add-ons ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

19. My income level affects whether I purchase add-on items ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

20. I buy add-on items when they are offered as combo deals ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

21. I prefer ordering add-ons to complete my meal experience ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

22. Which platform provides better personalized recommendations?

- ☐ Zomato
- ☐ Swiggy
- ☐ Both same

23. Which platform encourages more add-on purchases?

- ☐ Zomato
- ☐ Swiggy
- ☐ Both same

24. Overall, personalization features influence my add-on buying behaviour ?

- ☐ Strongly Disagree
- ☐ Disagree
- ☐ Neutral
- ☐ Agree
- ☐ Strongly Agree

References

- Ahn, J., & Kwon, H. (2025). AI-driven personalized recommendations and consumer purchase intentions in functional foods. *Journal of Consumer Behaviour*, 11(4), 333–349.
- Hallikainen, H., Laukkanen, T., & Törhönen, M. (2022). Consequences of personalized product recommendations in digital commerce. *Journal of Retailing and Consumer Services*, 64, 102000.
- Li, X., Wang, Y., & Zhang, H. (2024). Personalization algorithms and consumer decision-making in mobile commerce platforms. *Electronic Commerce Research and Applications*, 62, 101325.
- Prasetyo, H., Nugroho, A., & Putri, S. (2026). Consumer purchasing behavior of online food delivery application users. *International Journal of Consumer Studies*, 17(5), 168–182.
- Sánchez, C. N., et al. (2023). Recommendation system for a delivery food application based on number of orders. *Applied Sciences*, 13(4), 2299.
- Shorbaji, M. F. (2025). AI-enabled mobile food-ordering apps and customer experience: A systematic review. *Future Internet*, 20(3), 156.
- Tandon, A., Kaur, P., Bhatt, Y., Mäntymäki, M., & Dhir, A. (2021). Why do people purchase from food delivery apps? A consumer value perspective. *Journal of Consumer Behaviour*, 20(3), 517–534.
- Verhoef, P. C., Kannan, P. K., & Inman, J. J. (2023). From multi-channel retailing to omnichannel personalization: Understanding the role of data analytics in customer engagement. *Journal of Retailing*, 99(1), 1–18.
- Consumer buying behavior towards online food delivery apps. (2025). *International Journal of Consumer Research*, 15(2), 50–67.
- Impact of personalized advertisements on food delivery decisions. (2024). *ShodhKosh Journal of Social Sciences*, 10(12), 75–93.
- A study of consumer behaviour towards online food delivery. (2019). *Our Heritage*, 67(2), 36–49.
- Deloitte. (2024). *Digital consumer trends report*. Retrieved from <https://www2.deloitte.com>
- McKinsey & Company. (2023). *The value of getting personalization right or wrong is multiplying*. Retrieved from <https://www.mckinsey.com>
- Statista. (2024). *Online food delivery market outlook*. Retrieved from <https://www.statista.com>
- Dynamic Yield. (2023). *Online grocery personalization insights*. Retrieved from <https://www.dynamicyield.com>

Prasetyo, H., Nugroho, A., & Putri, S. (2026). Consumer purchasing behavior of online food delivery application users. *International Journal of Consumer Studies*, 17(5), 168–182.

Shorbaji, M. F. (2025). AI-enabled mobile food-ordering apps and customer experience: A systematic review. *Future Internet*, 20(3), 156.

Zhao, L. (2025). Understanding the influence of personalized recommendations on consumers' purchase intentions. *Journal of Digital Marketing Research*, 20(1), 32–50.

Ahn, J., & Kwon, H. (2025). AI-driven personalized recommendations and consumer purchase intentions in functional foods. *Journal of Consumer Behaviour*, 11(4), 333–349.

Consumer buying behavior towards online food delivery apps. (2025). *International Journal of Consumer Research*, 15(2), 50–67.

Impact of personalized advertisements on food delivery decisions. (2024). *ShodhKosh Journal of Social Sciences*, 10(12), 75–93.

Li, X., Wang, Y., & Zhang, H. (2024). Personalization algorithms and consumer decision-making in mobile commerce platforms. *Electronic Commerce Research and Applications*, 62, 101325.

Verhoef, P. C., Kannan, P. K., & Inman, J. J. (2023). From multi-channel retailing to omnichannel personalization: Understanding the role of data analytics in customer engagement. *Journal of Retailing*, 99(1), 1–18.

Sánchez, C. N., et al. (2023). Recommendation system for a delivery food application based on number of orders. *Applied Sciences*, 13(4), 2299.

Hallikainen, H., Laukkanen, T., & Törhönen, M. (2022). Consequences of personalized product recommendations in digital commerce. *Journal of Retailing and Consumer Services*, 64, 102000.

Tandon, A., Kaur, P., Bhatt, Y., Mäntymäki, M., & Dhir, A. (2021). Why do people purchase from food delivery apps? A consumer value perspective. *Journal of Consumer Behaviour*, 20(3), 517–534.

A study of consumer behaviour towards online food delivery. (2019). *Our Heritage*, 67(2), 36–49.

https://www.researchgate.net/publication/362761089_A_Study_of_Consumer_behaviour_towards_online_food_delivery

<https://www.mdpi.com/2076-3417/13/4/2299>

<https://www.sciencedirect.com/science/article/pii/S0969698922001813>

<https://onlinelibrary.wiley.com/doi/10.1002/cb.1898>

<https://www.sciencedirect.com/journal/electronic-commerce-research-and-applications>

<https://www.sciencedirect.com/journal/journal-of-retailing>

<https://www.mdpi.com/journal/futureinternet>

<https://www.granthaalayahpublication.org>

<https://www.mckinsey.com/capabilities/growth-marketing-and-sales/our-insights/the-value-of-getting-personalization-right-or-wrong-is-multiplying>

<https://www2.deloitte.com/global/en/pages/technology-media-and-telecommunications/articles/digital-consumer-trends.html>

<https://www.statista.com/outlook/dmo/online-food-delivery>

<https://www.dynamicsyield.com/article/online-grocery-personalization/>